

Focus on Web Technology



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1. Tools for Creating a Website

- Text editors
 - Similar to a word processing program, but lacks most text formatting features
 - Operating systems typically include a text editor
 - A code editor is a type of text editor that contains additional features to help web developers write code

Tools for Creating a Website

The diagram illustrates three different text editors used for creating a website, each displaying the same HTML code. The code is for an index.html file and includes a body style, a comment, a heading, a paragraph, a list of links, and a video link.

Notepad, in Windows

```
<body style="page-background" onload=showDateAndTime() >
<!-- Page content begins here -->

<h1>Web Development</h1>

<p>Three technologies form the foundation for many web applications:</p>
<ol>
  <li>
    <a href="http://www.w3.org/html/logo" target="_blank">HTML5</a>
    specifies the structure of content displayed on a web page;
  </li>
  <li>
    <a href="http://www.w3.org/Style/Examples/011/firstcss"
      target="_blank">CSS</a>
    (cascading style sheets) describes the design and appearance
    of information on a webpage; and
  </li>
  <li>
    <a href="http://javascript.com" target="_blank">JavaScript</a>
    allows users to interact with a webpage's content.
  </li>
</ol>

<p><a href="video.html">Watch a video</a> about web development.</p>

<h2>HTML5</h2>
```

TextEdit, in MacOS

```
<body style="page-background" onload=showDateAndTime() >
<!-- Page content begins here -->

<h1>Web Development</h1>

<p>Three technologies form the found
<ol>
  <li>
    <a href="http://www.w3.org/ht
    of content displayed on a webp
  </li>
  <li>
    <a href="http://www.w3.org/St
    (cascading style sheets) desc
    webpage; and
  </li>
  <li>
    <a href="http://javascript.co
    with a webpage's content.
  </li>
</ol>

<p><a href="video.html">Watch a video</a> about web development.</p>

<h2>HTML5</h2>

```

Brackets text editor

```
<body style="page-background" onload=showDateAndTime() >
<!-- Page content begins here -->

<h1>Web Development</h1>

<p>Three technologies form the foundation for many web applications:</p>
<ol>
  <li>
    <a href="http://www.w3.org/html/logo" target="_blank">HTML5</a>
    specifies the structure of content displayed on a web page;
  </li>
  <li>
    <a href="http://www.w3.org/Style/Examples/011/firstcss"
      target="_blank">CSS</a>
    (cascading style sheets) describes the design and appearance
    of information on a webpage; and
  </li>
  <li>
    <a href="http://javascript.com" target="_blank">JavaScript</a>
    allows users to interact with a webpage's content.
  </li>
</ol>

<p><a href="video.html">Watch a video</a> about web development.</p>

<h2>HTML5</h2>
```

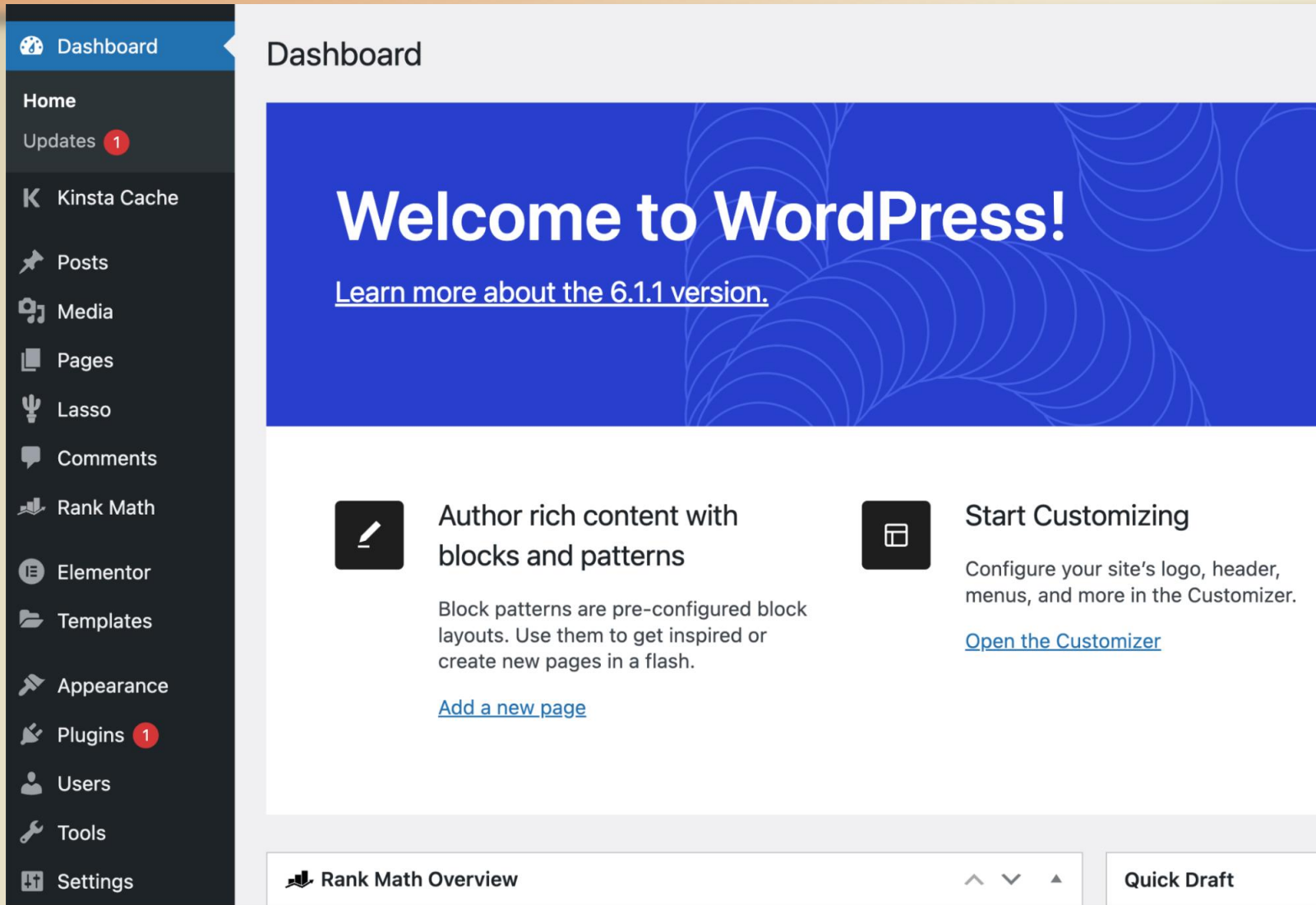
Tools for Creating a Website

- Content Management Systems
 - Enables and manages the publishing, modification, organization, and access of various forms of documents and other files on a network or the web
 - Web developer creates a theme, and one or more website content administrators enters the content

Example of CMS

 WordPress	 Drupal	 Joomla!
 Magento an Adobe Company	 Wix	 Squarespace
 Shopify	 TYPO3	 concrete CMS
 Sitecore	 Webflow	 Kentico

Tools for Creating a Website



The image shows the WordPress 6.1.1 Dashboard. On the left is a dark sidebar menu with icons and labels for various site management tools. The main content area has a light blue header with the word 'Dashboard'. Below this is a large blue banner with the text 'Welcome to WordPress!' and a link to learn more about the 6.1.1 version. The main area contains two white boxes with icons and text: 'Author rich content with blocks and patterns' and 'Start Customizing'. At the bottom, there is a 'Rank Math Overview' section and a 'Quick Draft' button.

Dashboard

Welcome to WordPress!

[Learn more about the 6.1.1 version.](#)

**Author rich content with blocks and patterns**

Block patterns are pre-configured block layouts. Use them to get inspired or create new pages in a flash.

[Add a new page](#)

**Start Customizing**

Configure your site's logo, header, menus, and more in the Customizer.

[Open the Customizer](#)

 **Rank Math Overview**

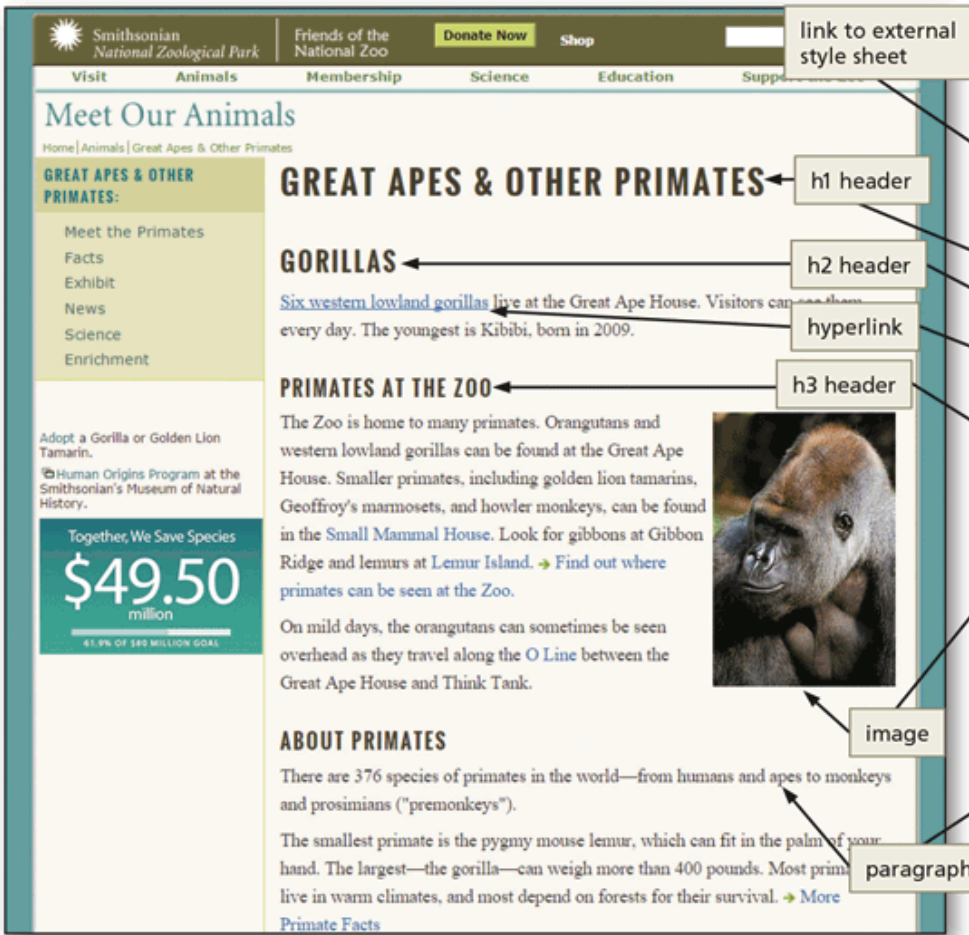
Quick Draft

- Dashboard
- Home
- Updates **1**
- K Kinsta Cache
- Posts
- Media
- Pages
- Lasso
- Comments
- Rank Math
- Elementor
- Templates
- Appearance
- Plugins **1**
- Users
- Tools
- Settings

2. Website Technologies

- Hypertext Markup Language (**HTML**) uses a set of codes called **tags** to format documents for display in a browser
- A complementary technology called **cascading style sheets** (**CSS**) contains specifications for the fonts, colors, layout, and placement of these HTML elements on a webpage
- **JavaScript** is a programming language for creating programs that a browser can run to generate content for a website

Website Technologies



link to external style sheet

h1 header

h2 header

hyperlink

h3 header

image

paragraph

```
1 <!DOCTYPE html
2 PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
3 "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
4 <html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en"><!-- Inst
5 codeOutsideHTMLIsLocked="false" -->
6 <head>
7
8 <title>Great Apes and Other Primates - National Zoo</title>
9
10 <link href="https://fonts.googleapis.com/css?family=Oswald:400,300,700" rel=
11 />
12 </head>
13 <body>
14
15 <h1>Great Apes & Other Primates</h1>
16
17 <h2>Gorillas</h2>
18
19 <p>
20 <a href="/Animals/Primates/MeetPrimates/MeetGorillas/">Six western lowland
21 youngest is Kibibi, born in 2009.
22 </p>
23
24 <h3>Primates at the Zoo </h3>
25
26 <a href="http://www.flickr.com/photos/nationalzoo/3707340844/" title="Goril
27 
29
30 The Zoo is home to many primates. Orangutans and western lowland gorillas
31 tamarins, Geoffroy's marmosets, and howler monkeys, can be found in the
32 Gibbon Ridge and lemurs at <a href="/animals/primates/meetprimates/meetlemur
33 src="/admin/images/icons/img_arrowdk.gif" alt="" width="15" height="11" />F
34 </p>
35
36 On mild days, the orangutans can sometimes be seen overhead as they travel
37 Ape House and Think Tank. </p>
38 <h3>About Primates</h3>
39
40 <p>There are 376 species of primates in the world&#8212;from humans and apes
41 </p>
42
43 The smallest primate is the pygmy mouse lemur, which can fit in the palm of
44 pounds. Most primates live in warm climates, and most depend on forests for
45 src="/admin/images/icons/img_arrowdk.gif" alt="" width="15" height="11" />M
46 </p>
47
48 </body>
49 </html>
```


Website Technologies

What is HTML?

- HTML stands for **Hyper Text Markup Language**
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements** tell the browser how to display the content
- HTML elements** label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

Ref: https://www.w3schools.com/html/html_intro.asp

Website Technologies

What is an HTML Element?

An HTML element is defined by a start tag, some content, and an end tag:

`<tagname> Content goes here... </tagname>`

The HTML **element** is everything from the start tag to the end tag:

`<h1>My First Heading</h1>`

`<p>My first paragraph.</p>`

Start tag	Element content	End tag
<code><h1></code>	My First Heading	<code></h1></code>
<code><p></code>	My first paragraph.	<code></p></code>
<code>
</code>	<i>none</i>	<i>none</i>

Ref: https://www.w3schools.com/html/html_intro.asp

Website Technologies

A Simple HTML Document

Example

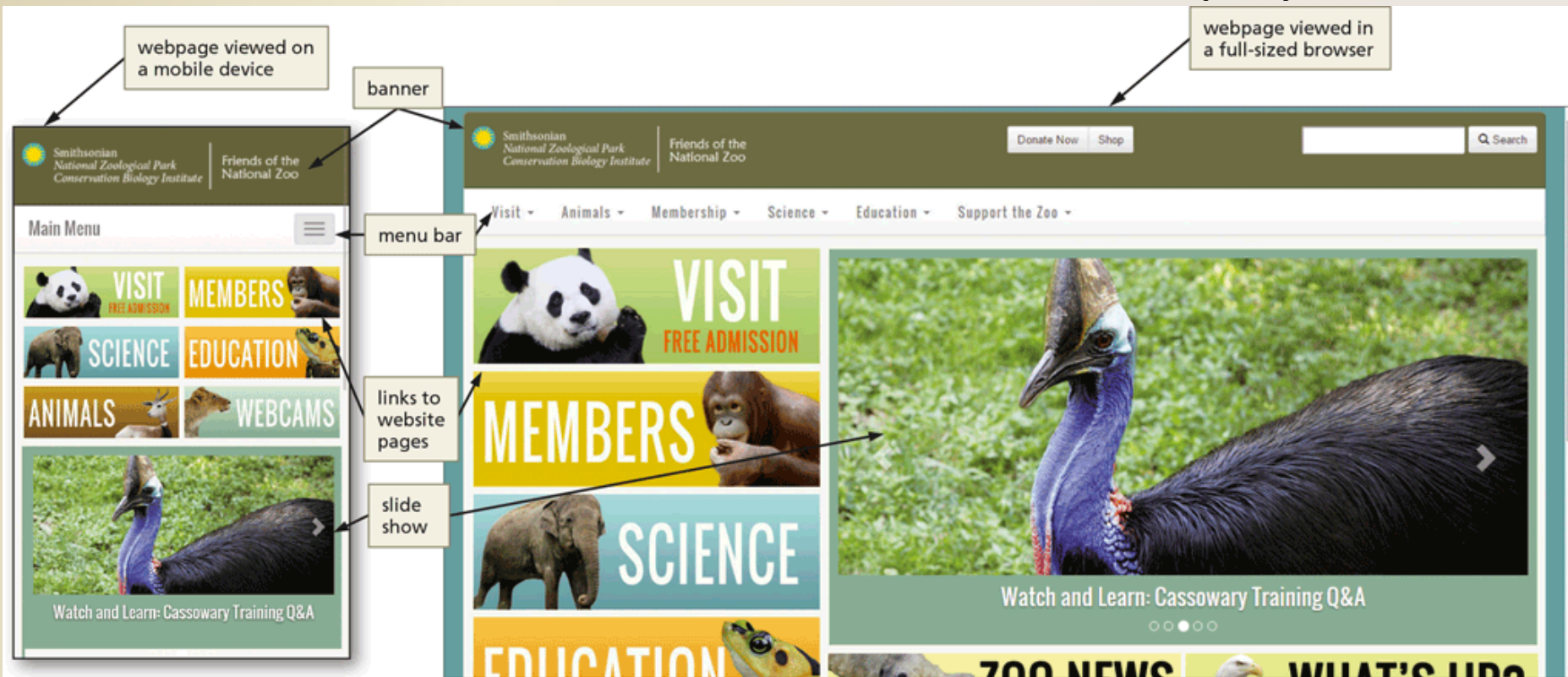
```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

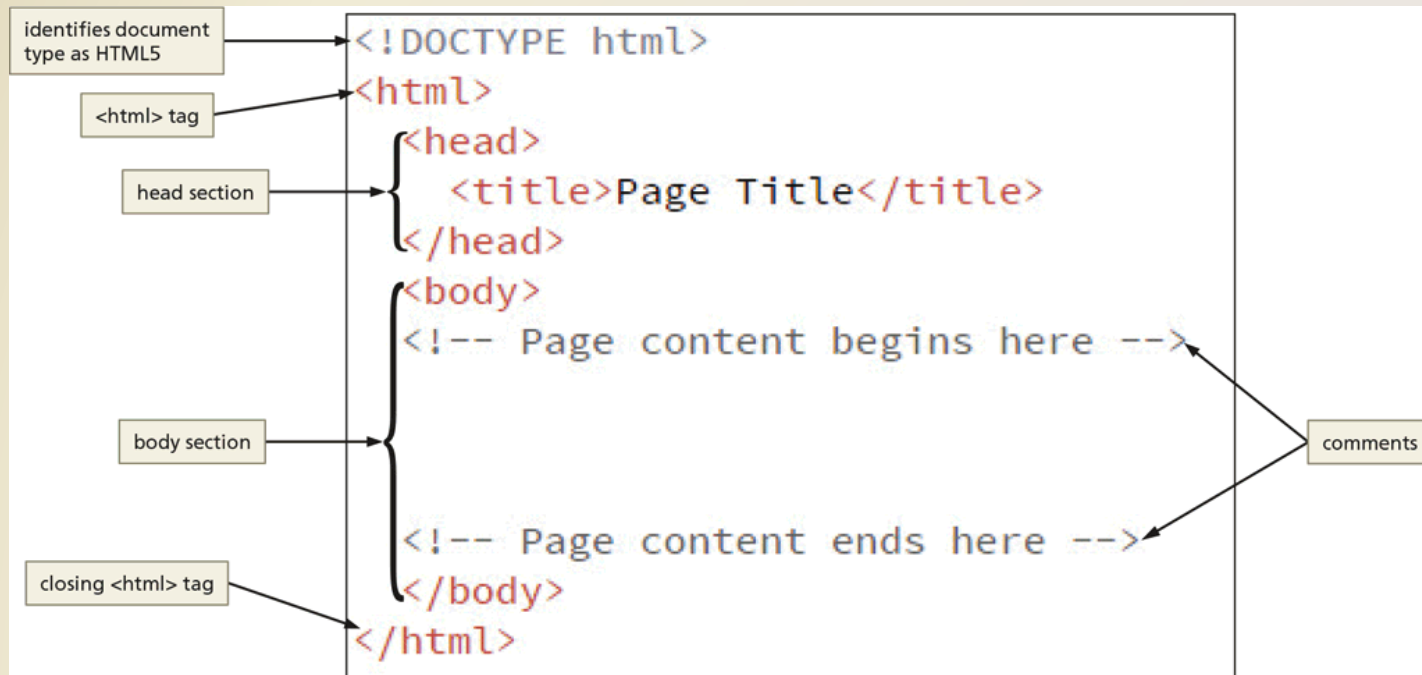
Website Technologies

- **Responsive webpages** automatically adjust the size of their content to display appropriately relative to the size of the screen of the device on which it is displayed



3. Structure of a Webpage

- A webpage's source code contains text marked up with **HTML tags** that instruct a browser how to display that content



Structure of a Webpage

- The World Wide Web Consortium (W3C) oversees the specification of HTML Standards
- The W3C provides a free, online HTML5 validator application to ensure that a webpage's HTML tags follow the specifications, or rules, for HTML5

4. Creating the index.html File

- Run the text editor of your choice
- Navigate to and open the template.html file
- If necessary, enable the word wrap feature so that you can view all the webpage text without scrolling horizontally
- Save the file using the file name, index.html. Do not exit the text editor

Creating the index.html File

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Page Title</title>
5   </head>
6   <body>
7     <!-- Page content begins here -->
8
9
10
11     <!-- Page content ends here -->
12   </body>
13 </html>
```

webpage structure

4.1 Adding the Webpage Title

- Select the text, **Page Title**, that appears between the `<title>` and `</title>` tags
- Type Mark's Web Development Page as the title. Replace the name, Mark, with your first name

4.2 Headings

- Headings **indicate the different sections** of a webpage. HTML supports six levels of headings, which are identified by the following tags: **<h1>**, **<h2>**, **<h3>**, **<h4>**, **<h5>**, and **<h6>**

Headings

```
<html>
  <head>
    <title>Headings</title>
  </head>
  <body>
    <h1>Heading level 1</h1>
    <h2>Heading level 2</h2>
    <h3>Heading level 3</h3>
    <h4>Heading level 4</h4>
    <h5>Heading level 5</h5>
    <h6>Heading level 6</h6>
  </body>
</html>
```

The diagram illustrates the mapping between HTML heading tags and their visual output. Arrows point from the closing tags in the code to the corresponding heading levels:

- `</h1>` points to **Heading level 1**
- `</h2>` points to **Heading level 2**
- `</h3>` points to **Heading level 3**
- `</h4>` points to **Heading level 4**
- `</h5>` points to **Heading level 5**
- `</h6>` points to **Heading level 6**

4.3 Paragraphs

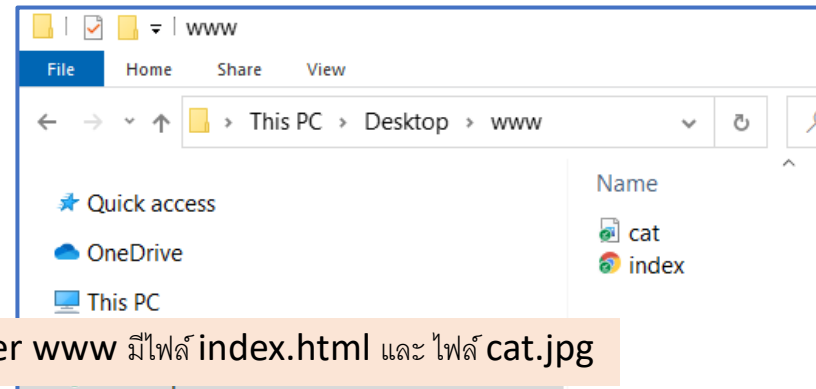
- The `<p>` and `</p>` tags are used to identify the beginning and ending of Paragraphs
- If you have several paragraphs of text on your webpage, these tags will inform the browser to insert additional line spacing above and below the paragraph so that the text is easier to read when displayed in the browser
 - The browser ignores line breaks and line spacing in the HTML file, so it is important to properly define the paragraphs using the `<p>` and `</p>` tags

4.4 Images

- Images can be either photos or graphics
- Images always are stored as separate files, and references to the images appear in the HTML code using the `<img>` tag
- Common attributes for the `<img>` tag describe the location of an image file, alternate text for the image, and a style that indicates how to position the image
- ``

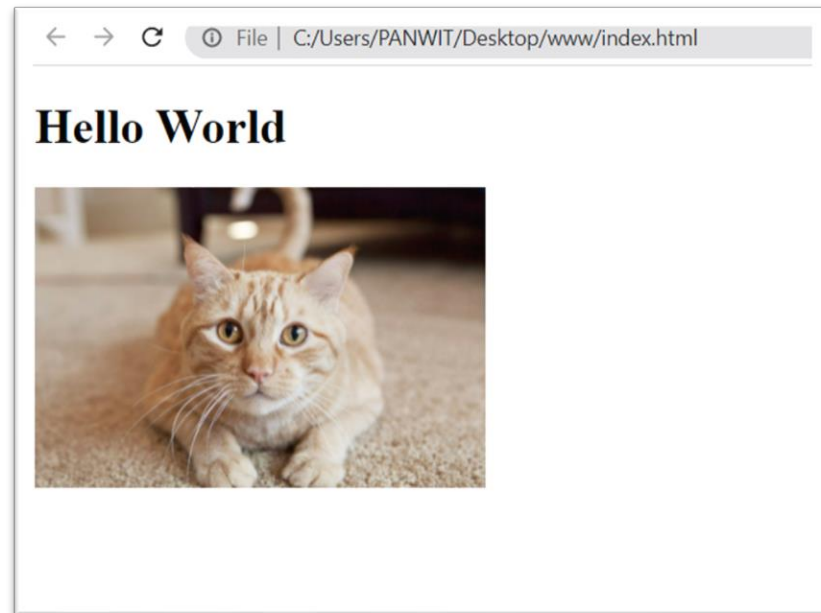
การอ้างอิงตำแหน่งไฟล์รูป ใน html

- File นั้นๆ อยู่ใน folder เดียวกัน
เช่น ไฟล์รูป และไฟล์ index.html
อยู่ใน **folder** เดียวกัน



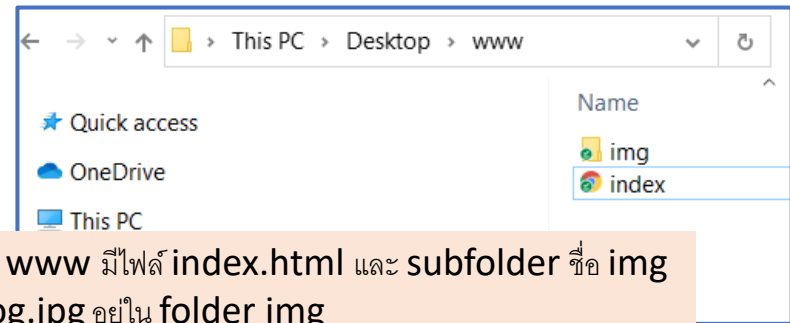
ไฟล์ index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Website Title</title>
  </head>
  <body>
    <h1>Hello World</h1>
    
  </body>
</html>
```



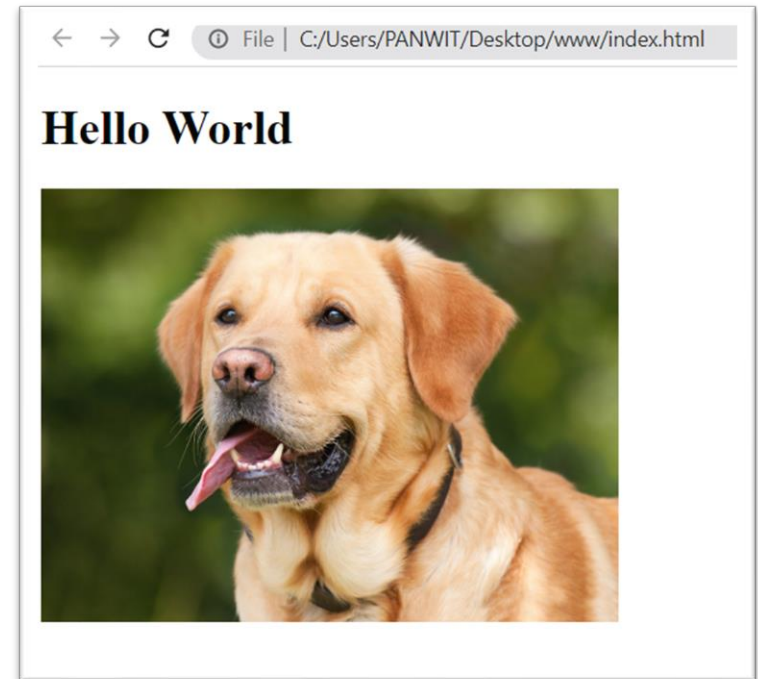
การอ้างอิงตำแหน่งไฟล์รูป ใน html

- File นั้นๆ อยู่คนละ folder เช่น ไฟล์รูป และไฟล์ index.html อยู่คนละ **folder**



ไฟล์ index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Website Title</title>
  </head>
  <body>
    <h1>Hello World</h1>
    
  </body>
</html>
```



4.5 Links

- A link, or **hyperlink**, can be text or an image in a webpage that a user clicks to navigate to another webpage, download a file, or perform another action, such as running an email app and addressing an email message

HYPERLINK WITH ABSOLUTE REFERENCE

```
<a href="http://www.w3.org" target="_blank">HTML5</a>
```

HYPERLINK WITH RELATIVE REFERENCE

```
<a href="video.html"> Watch a video</a>
```

4.6 Unordered and Ordered Lists

Unordered List

```
<ul>  
  <li>HTML5</li>  
  <li>CSS</li>  
  <li>JavaScript</li>  
</ul>
```

- HTML5
- CSS
- JavaScript

Ordered List

```
<ol>  
  <li>HTML5</li>  
  <li>CSS</li>  
  <li>JavaScript</li>  
</ol>
```

1. HTML5
2. CSS
3. JavaScript

Basic HTML

ไฟล์ index.html

แสดงผลที่ Web browser

```
index - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html>
<head>
<title>My First HTML</title>
</head>
<body onload=showDateAndTime() >

<h1>My First Heading</h1>

<p>My First Paragraph</p>

<p>HTML links are defined with the a tag:</p>

<a href="https://www.w3schools.com">This is a link</a>

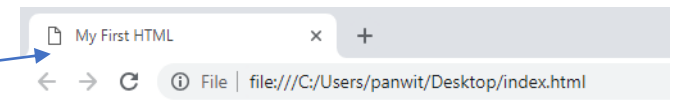
<h2>An Unordered HTML List</h2>

<ul>
<li>Coffee</li>
<li>Tea</li>
<li>Milk</li>
</ul>

<h2>An Ordered HTML List</h2>

<ol>
<li>Coffee</li>
<li>Tea</li>
<li>Milk</li>
</ol>

</body>
</html>
```



My First Heading

My First Paragraph

HTML links are defined with the a tag:

[This is a link](https://www.w3schools.com)

An Unordered HTML List

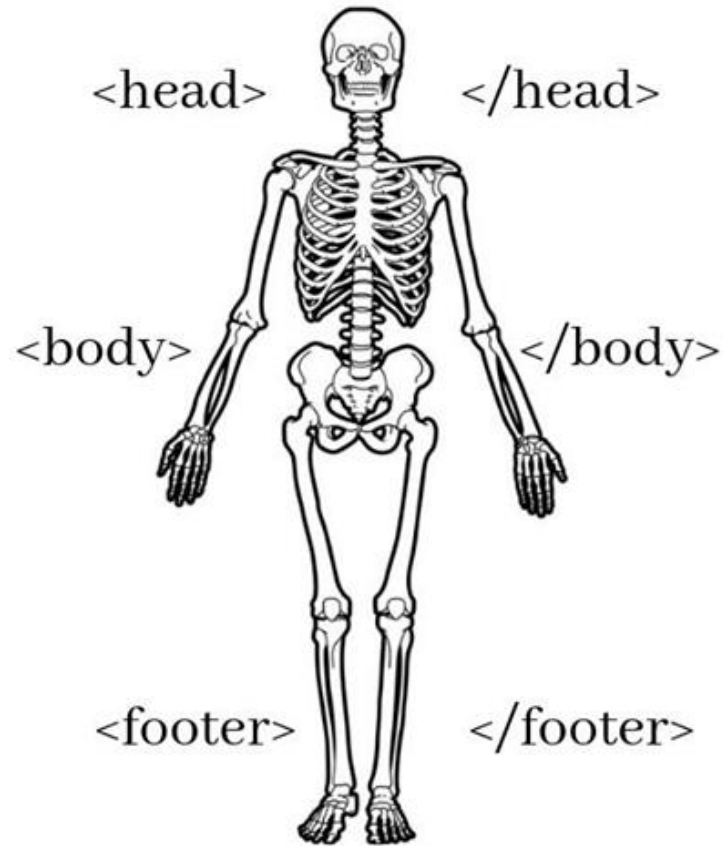
- Coffee
- Tea
- Milk

An Ordered HTML List

1. Coffee
2. Tea
3. Milk

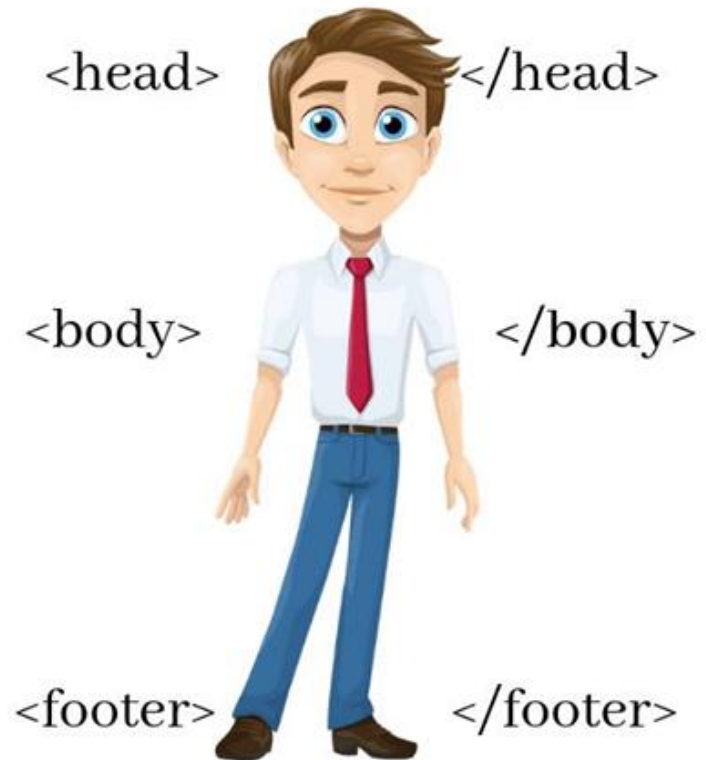
HTML Vs CSS

HTML



Structural Layer

HTML + CSS



Presentational Layer

5. HTML with CSS (Cascading Style Sheet)

ไฟล์ index.html

Internal CSS

```
<head>
<title>My First HTML</title>
<style>
body {
  background-color: lightblue;
}

h1 {
  font-family: verdana;
  color: #ff00ff;
  text-align: center;
}

h2 {
  font-family: cursive;
  font-size: 15px;
  font-style: italic;
}
</style>
</head>
<body>

<h1>My First Heading</h1>

<p>My first paragraph.</p>

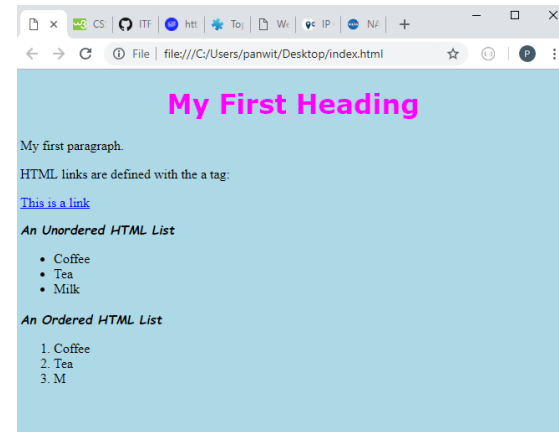
<p>HTML links are defined with the a tag:</p>

<a href="https://www.w3schools.com">This is a link</a>

<h2>An Unordered HTML List</h2>

<ul>
```

แสดงผลที่ Web browser



- Embedded styles, which define styles in the **<head>** section of the index.html document, apply to the entire webpage on which they are defined

5.1 Applying Styles with CSS

- Although it is possible to customize these elements of webpages using HTML, CSS makes it easier to specify the appearance of similar elements in the same webpage or same website
- Embedded styles, which **define styles in the <head> section of the index.html document, apply to the entire webpage** on which they are defined

```
<style>
  h1 {
    font-family:sans-serif;
    color:navy;
    font-style:italic;
  }
  h2 {
    font-family:cursive;
    background-color: navy;
    color:papayawhip;
  }
  p {
    font-family:sans-serif;
    color:rgb(56,0,0);
  }
  ol {
    font-family:sans-serif;
    color:rgb(56,0,0);
  }
  body {
    background-color:#bbccff;
  }
  .fancy {
    font-weight:bold;
    color:red;
    font-style:italic;
  }
</style>
```

5.2 Adding CSS to the index.html File

styles entered in
head section of
index.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Mark's Web Development Page</title>
5   <style>
6     h1 {
7       font-family:sans-serif;
8       color:navy;
9       font-style:italic;
10    }
11    h2 {
12      font-family:cursive;
13      background-color:navy;
14      color:papayawhip;
15    }
16    p {
17      font-family:sans-serif;
18      color:rgb(56,0,0);
19    }
20    ol {
21      font-family:sans-serif;
22      color:rgb(56,0,0);
23    }
24    body {
25      background-color:#bbccff;
26    }
27    .fancy {
28      font-weight:bold;
29      color:red;
30      font-style:italic;
31    }
32  </style>
33 </head>
34 <body>
35 <!-- Page content begins here -->
36 <h1>Web Development</h1>
37
38 <p>Three technologies form the foundation for many web applications:</p>
39
40 <ol>
```

5.3 Styling HTML with CSS

- CSS can be added to HTML elements in 3 ways
 - Inline - by using the style attribute in HTML elements
 - Internal - by using a <style> element in the <head> section
 - External - by using an external CSS file

Inline

```
<h1 style="color:blue;">This is a Blue Heading</h1>
```



Internal

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1 {color: blue;}
p {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```



External

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="styles.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

```
body {
  background-color: powderblue;
}
h1 {
  color: blue;
}
p {
  color: red;
}
```

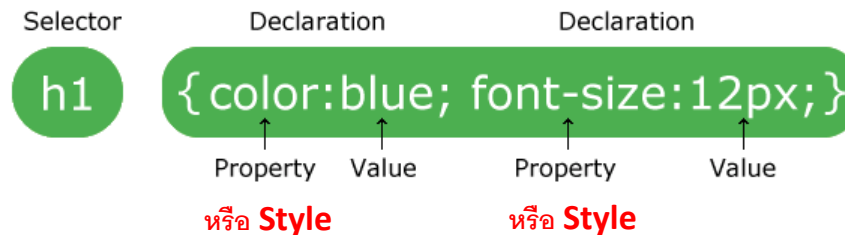
ไฟล์ style.css

ไฟล์ index.html



5.4 CSS Syntax

A CSS rule-set consists of a selector and a declaration block:



- The **selector** points to the **HTML element** you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS **property name** and a **value**, separated by a colon.
- A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

ที่มา https://www.w3schools.com/css/css_syntax.asp

CSS Selector

CSS selectors are used to "find" (or select) the HTML elements you want to style.

All CSS Simple Selector

1. The CSS **element** Selector

```
p {  
  text-align: center;  
  color: red;  
}
```

3. The CSS **class** Selector

```
.center {  
  text-align: center;  
  color: red;  
}
```

5. The CSS **group** Selector

```
h1, h2, p {  
  text-align: center;  
  color: red;  
}
```

2. The CSS **id** Selector

```
#para1 {  
  text-align: center;  
  color: red;  
}
```

4. The CSS **universal** Selector

```
* {  
  text-align: center;  
  color: blue;  
}
```

ที่มา https://www.w3schools.com/css/css_selectors.asp

Example of Style

ศึกษาเพิ่มเติมได้ที่

<https://www.w3schools.com/css/default.asp>

https://www.w3schools.com/css/css_font.asp

https://www.w3schools.com/css/css_text.asp

Table 3 Selected Styles

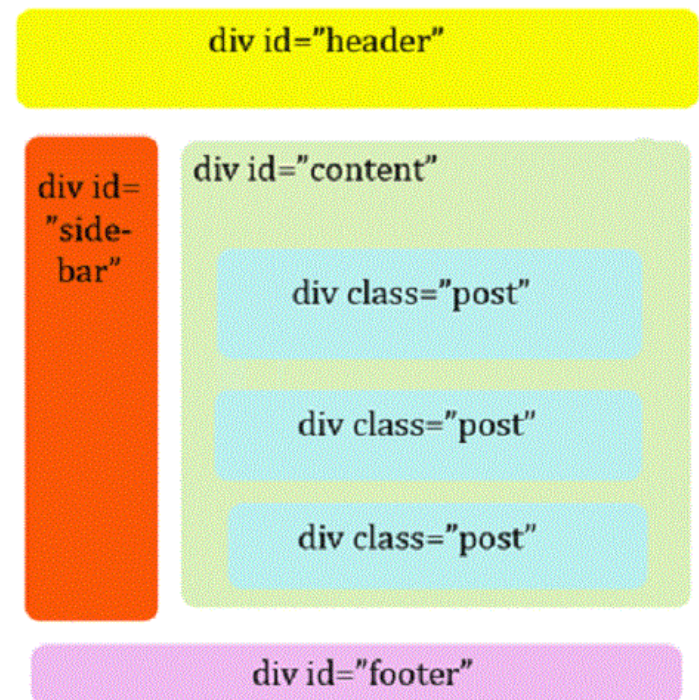
Style	Example	Description
background-color	background-color:yellow;	Specifies the background color of elements, such as <code><p></code> , <code><h1></code> , and <code><body></code>
background-image	background-image:url("images/stripes.jpg")	Sets the background image of a <code><body></code> , <code><p></code> , <code><h1></code> , and other elements to the file whose path is given in the <code>url()</code> function
border	border: 3 px red;	Specifies a 4-sided border that is 3 pixels thick
color	color:blue;	Colors can be a web color name, a hexadecimal value, such as <code>#0000FF</code> , or an rgb value, such as <code>rgb(0,0,255)</code> that specifies the red, green, and blue components of the color
float	float:left;	Specifies whether to place an element to the left or right relative to text; often used to position an image to the left or right of text.
font-family	font-family:serif;	Specifies the font for a paragraph, heading, or other text element; use specific font names, such as <code>times</code> , <code>arial</code> , and <code>courier</code> , or family names, such as <code>serif</code> , <code>sans-serif</code> , <code>cursive</code> , or <code>monospace</code>
font-size	font-size:10px;	Specifies font size in pixels
font-style	font-style:italic;	Specifies font style; use <code>normal</code> , <code>italic</code> , or <code>oblique</code> as the value for this style
font-weight	font-weight:bold;	Specifies the font weight for a paragraph, heading, or other text element; use <code>bold</code> for thick characters, or numeric values 100 through 900, in increments of 100; 400 is the same as <code>normal</code> , 700 is the same as <code>bold</code>
text-align	text-align:left;	Sets alignment for <code><h1></code> , <code><h2></code> , or <code><h3></code> tags; values can be <code>left</code> , <code>center</code> , or <code>right</code>

5.5 Tag <div>

div หรือ division คือ tag ประเภทหนึ่งของภาษา html

- นำมาใช้แบ่งส่วนต่างๆ ของข้อมูลออกจากกัน
- ใช้สำหรับครอบวัตถุที่ต้องการ เพื่อจัดรูปแบบต่างๆ ให้กับวัตถุในตำแหน่งนั้นๆ
- สามารถใส่ค่าไปตรงๆ ใน div โดยระบุ id หรือ class เพื่อใช้อ้างอิงกับการจัด CSS ได้ตามความเหมาะสม

An HTML page can only have one **unique id** apply to **one specific element**, while a **class** name can be **applied to multiple elements**.



```

<!DOCTYPE html>
<html>
<head>
<style>
/* Style the element with the id "myHeader" */
#myHeader {
  background-color: lightblue;
  color: black;
  padding: 40px;
  text-align: center;
}

/* Style all elements with the class name "city" */
.city {
  background-color: tomato;
  color: white;
  padding: 10px;
}
</style>
</head>
<body>

<h2>Difference Between Class and ID</h2>
<p>An HTML page can only have one unique id applied to one
specific element, while a class name can be applied to multiple
elements.</p>

<!-- A unique element -->
<h1 id="myHeader">My Cities</h1>

<!-- Multiple similar elements -->
<h2 class="city">London</h2>
<p>London is the capital of England.</p>

<h2 class="city">Paris</h2>
<p>Paris is the capital of France.</p>

<h2 class="city">Tokyo</h2>
<p>Tokyo is the capital of Japan.</p>

</body>
</html>

```

Difference Between Class and ID

An HTML page can only have one unique id applied to one specific element, while a class name can be applied to multiple elements.

My Cities

London

London is the capital of England.

Paris

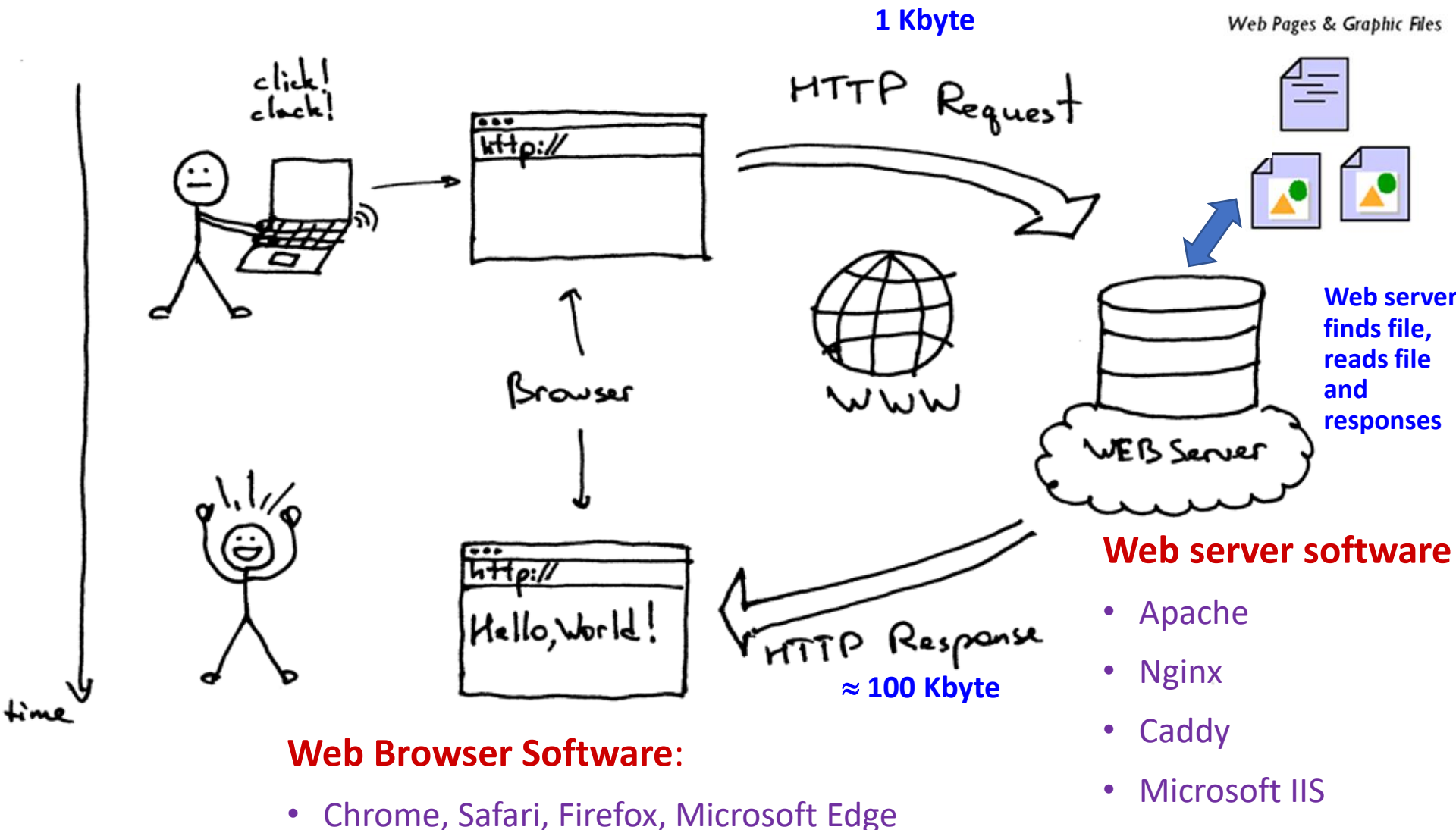
Paris is the capital of France.

Tokyo

Tokyo is the capital of Japan.

การใช้ id ใน CSS จะถูกกำหนดด้วยเครื่องหมาย #
การใช้ class ใน CSS จะถูกกำหนดด้วยเครื่องหมายจุด .

6. ภาพรวม web client-server



client

HTML
Parser



Rendered
Document



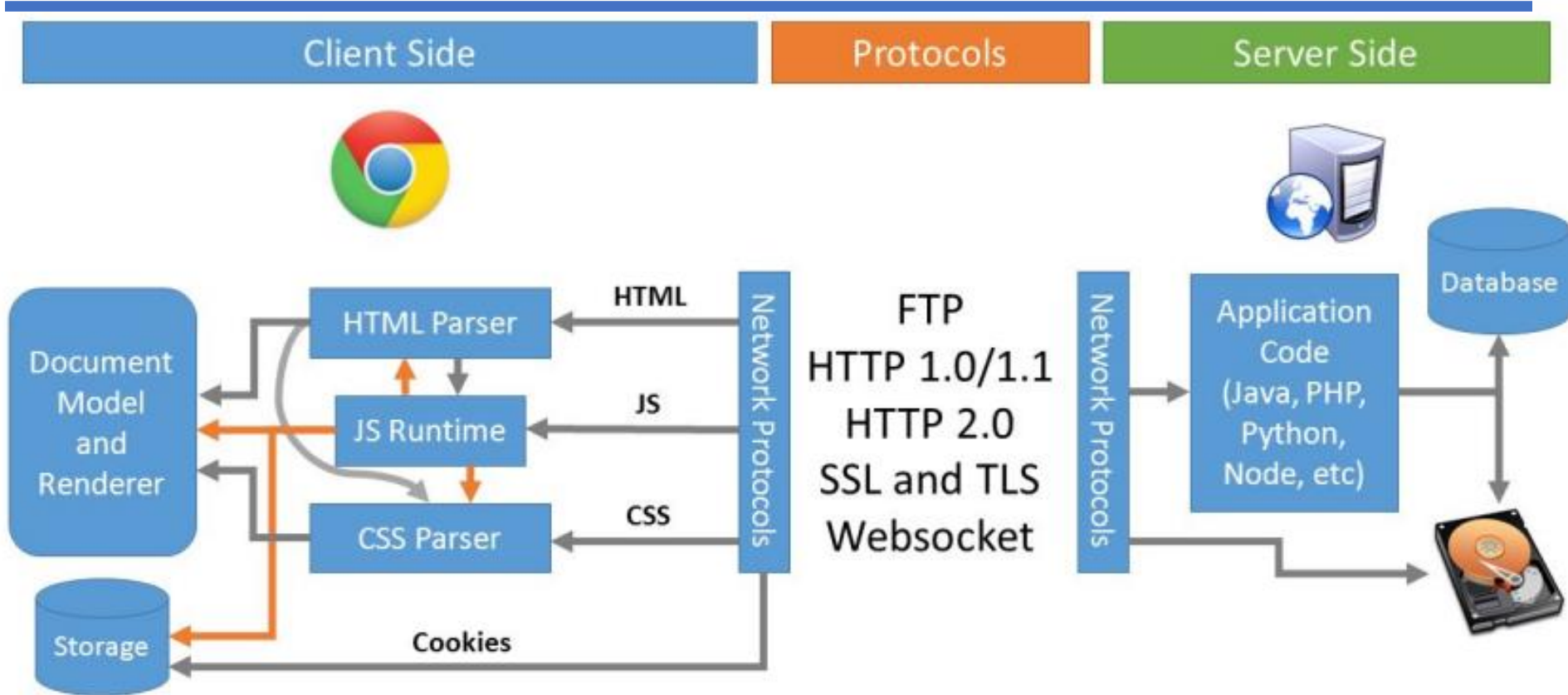
server

index.html

HTML Parser คือตัวแปลง code html ให้กลายเป็น User Interface แสดงผลให้เห็นบน web browser

HTML Parser ทำหน้าที่ render พวก tag ต่าง ๆ ของ code html ออกมาเป็น User Interface

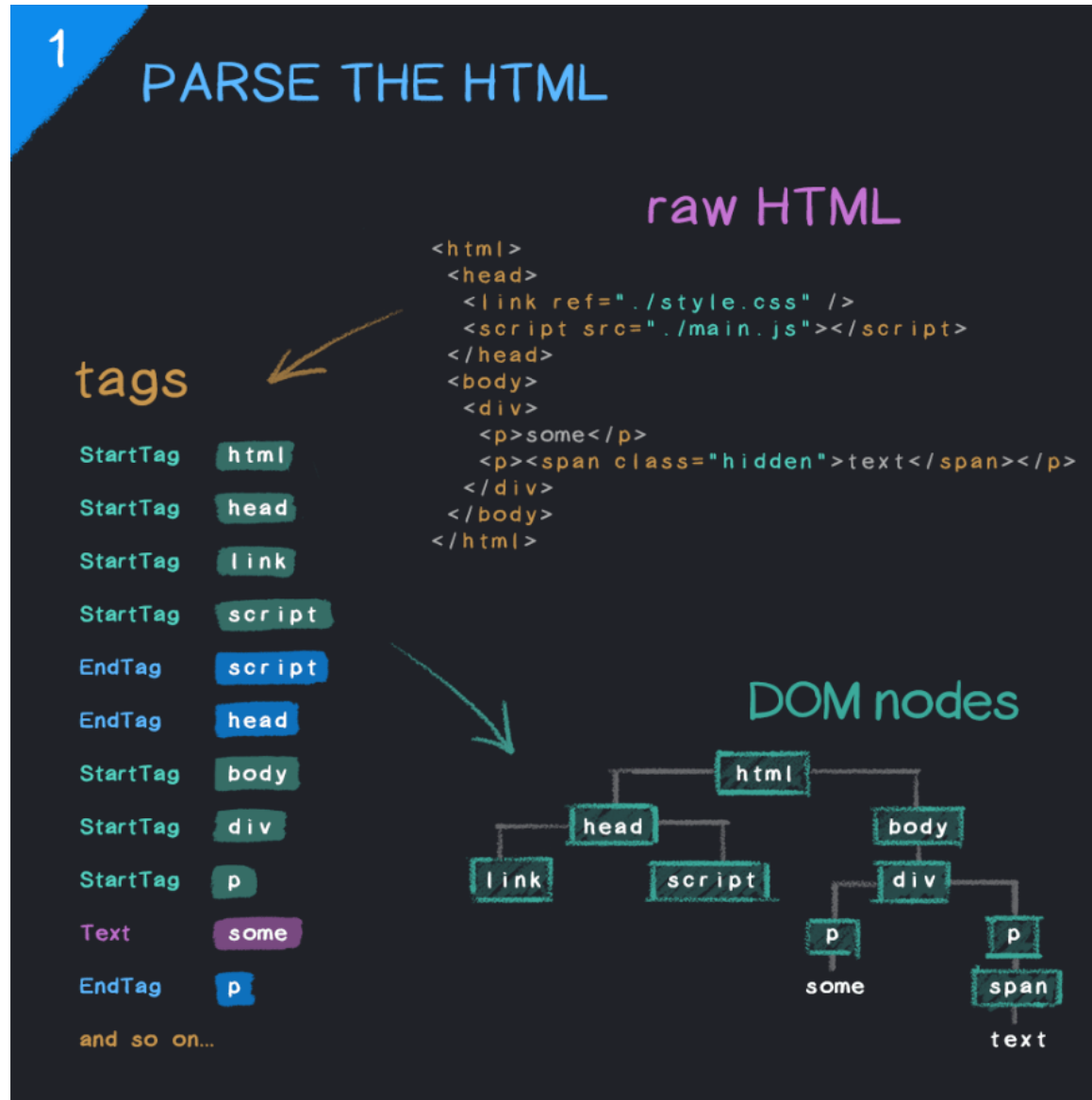
7. การทำงานของ Web Browser



ตัววิเคราะห์คำ แยกคำ ในประโยค

- **HTML Parser** : ทำหน้าที่ render พวก tag ต่าง ๆ ของ code html ออกมาเป็น DOM
 - **CSS Parser** : ส่วนไหนใน code html เป็น css ก็เข้า CSS Parser ออกมาเป็น CSSOM
 - **JS Runtime** : ถ้าใน code html มี JavaScript (js) มันก็จะต้องการ JS runtime environment ในการ run ให้ code js ทำงาน โดยสามารถใช้ js ทำการ อ่าน/แก้ไข html หรือจัดการ css ได้ ซึ่ง browser ทุกตัวบนโลกเดี๋ยวนี้นี้มี js runtime อยู่แล้ว
- JavaScript Runtime ทำหน้าที่เพื่ออ่าน code ภาษา java script และทำงานตามคำสั่ง เพื่อให้ OS ต่างๆ ทำงานได้

7. การทำงานของ Web Browser



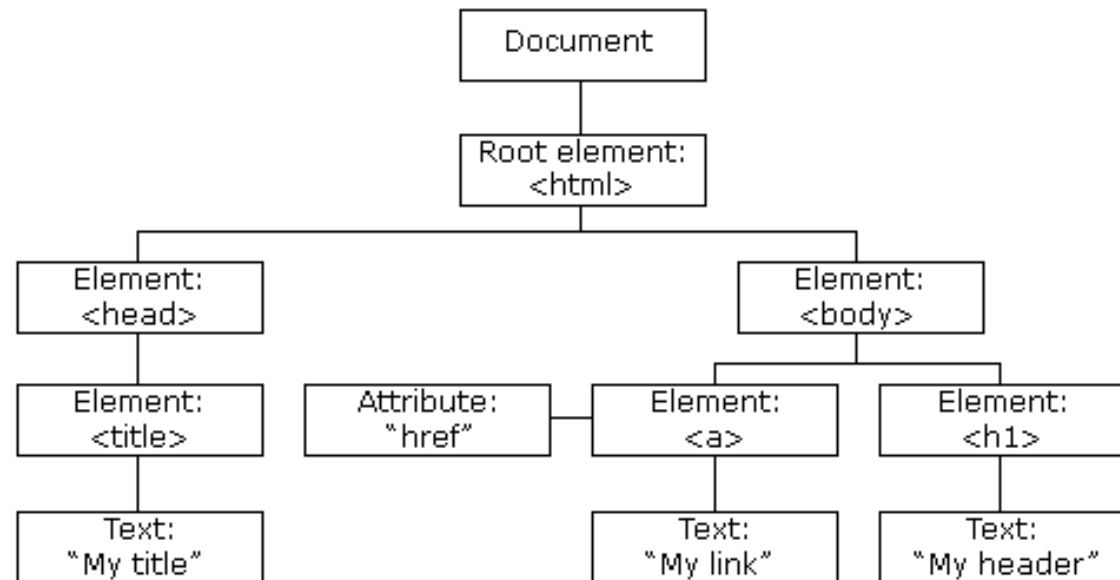
When a web page is loaded, the browser creates a Document Object Model of the page.

DOM หรือ Document Object Model คือโครงสร้างของไฟล์ ที่ทาง W3C กำหนดขึ้นเป็นมาตรฐาน

W3C ย่อมาจาก World Wide Web Consortium คือ องค์กรระหว่างประเทศที่ทำหน้าที่จัดระบบมาตรฐานที่ใช้งานบน www

```
1 <html> <!-- Root Element -->
2
3 <head>
4   <title>My title</title>
5 </head>
6
7 <body>
8   <a href="test.html">My link</a>
9   <h1>My header</h1>
10 </body>
11
12 </html>
```

index.html hosted with ❤ by GitHub



7. การทำงานของ Web Browser

2

FETCH EXTERNAL RESOURCES

CSS blocks rendering

JavaScript blocks parser...

unless!

defer waits until parser is finished

async executes as soon as it loads

7. การทำงานของ Web Browser

3

PARSE THE CSS AND BUILD THE CSSOM

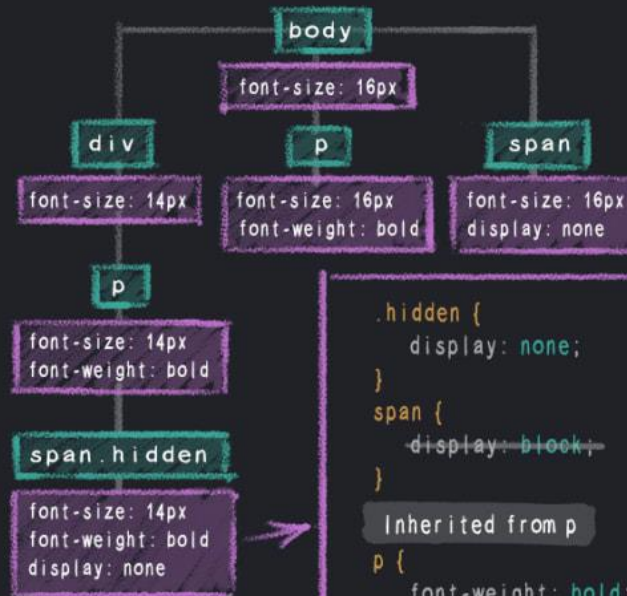
```
body {  
  font-size: 16px;  
}
```

```
div {  
  font-size: 14px;  
}
```

```
p {  
  font-weight: bold;  
}
```

```
.hidden {  
  display: none;  
}
```

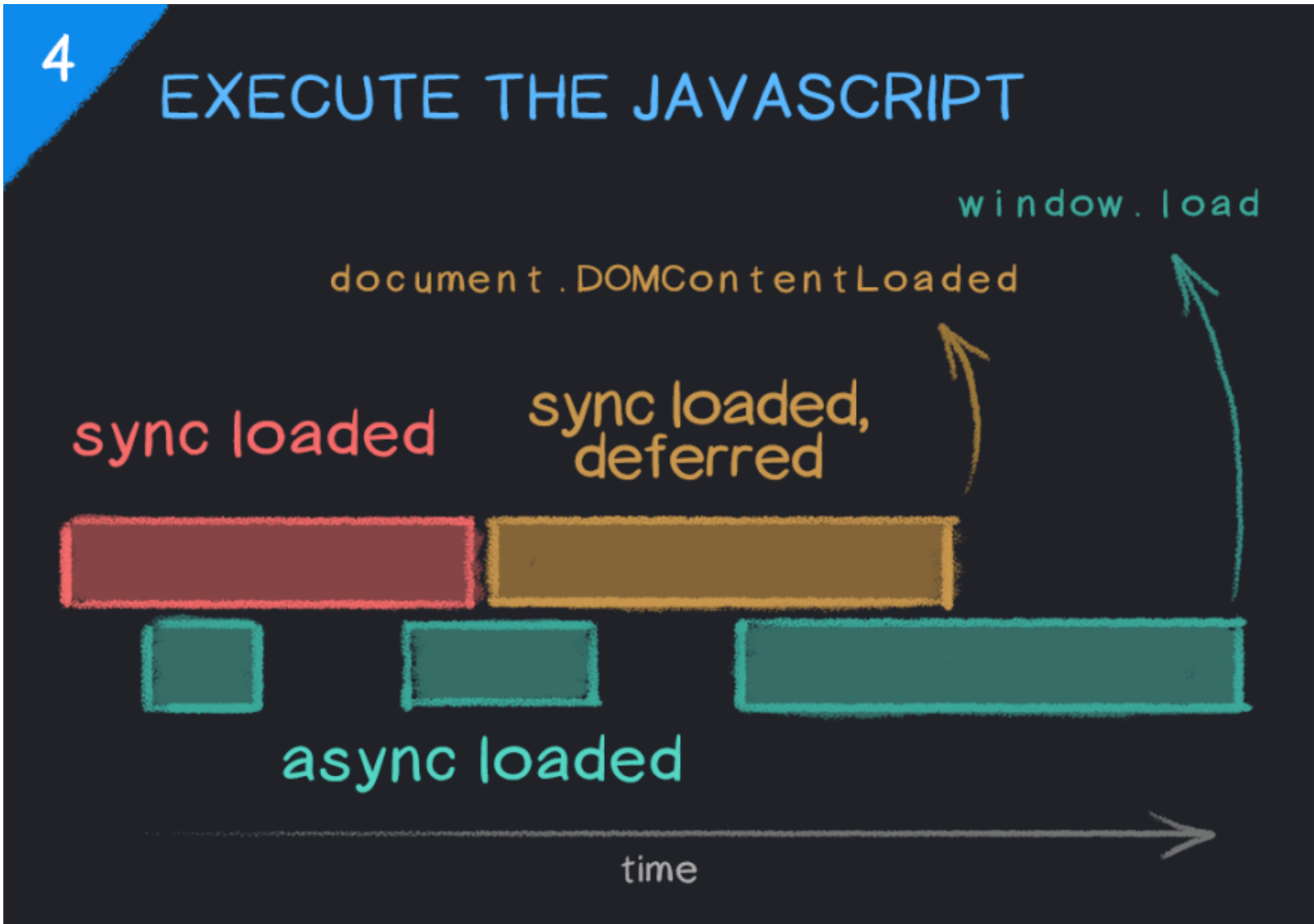
```
span {  
  display: block;  
}
```



```
.hidden {  
  display: none;  
}  
span {  
  display: block;  
}  
Inherited from p  
p {  
  font-weight: bold;  
}  
Inherited from div  
div {  
  font-size: 14px;  
}  
Inherited from body  
body {  
  font-size: 16px;  
}
```

Any CSS style can potentially be overridden or changed by some other style loaded on the page, which is why the CSSOM cannot be fully constructed until all of the page's stylesheets have been loaded.

7. การทำงานของ Web Browser

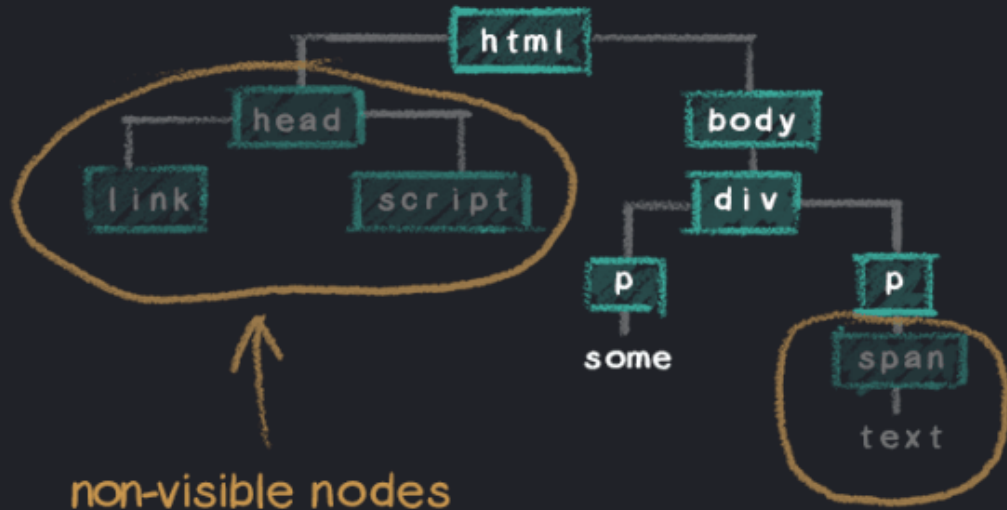


7. การทำงานของ Web Browser

5

MERGE DOM AND CSSOM TO CONSTRUCT THE RENDER TREE

```
body {  
  font-size: 16px;  
}  
  
div {  
  font-size: 14px;  
}  
  
p {  
  font-weight: bold;  
}  
  
span {  
  display: none;  
}
```



non-visible nodes

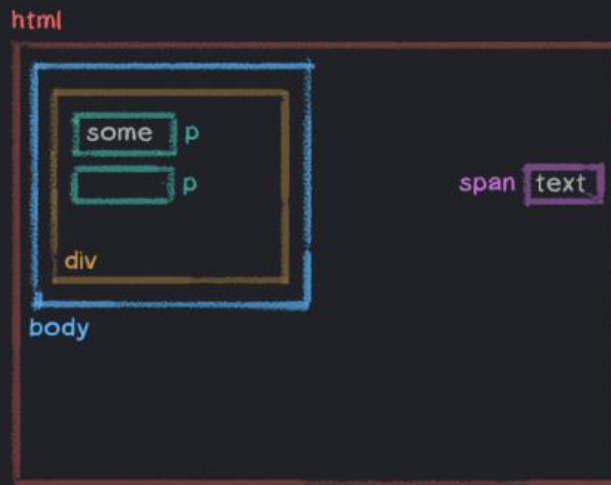
excluded from the render tree → "display: none"

7. การทำงานของ Web Browser

6

CALCULATE LAYOUT AND PAINT

```
<html>
<head>
  <link ref="./style.css" />
</head>
<body>
  <div>
    <p>some</p>
    <p><span>text</span></p>
  </div>
</body>
</html>
```



let's add some layout styles...

```
* {
  padding: 25px;
}

html {
  height: 100%;
}

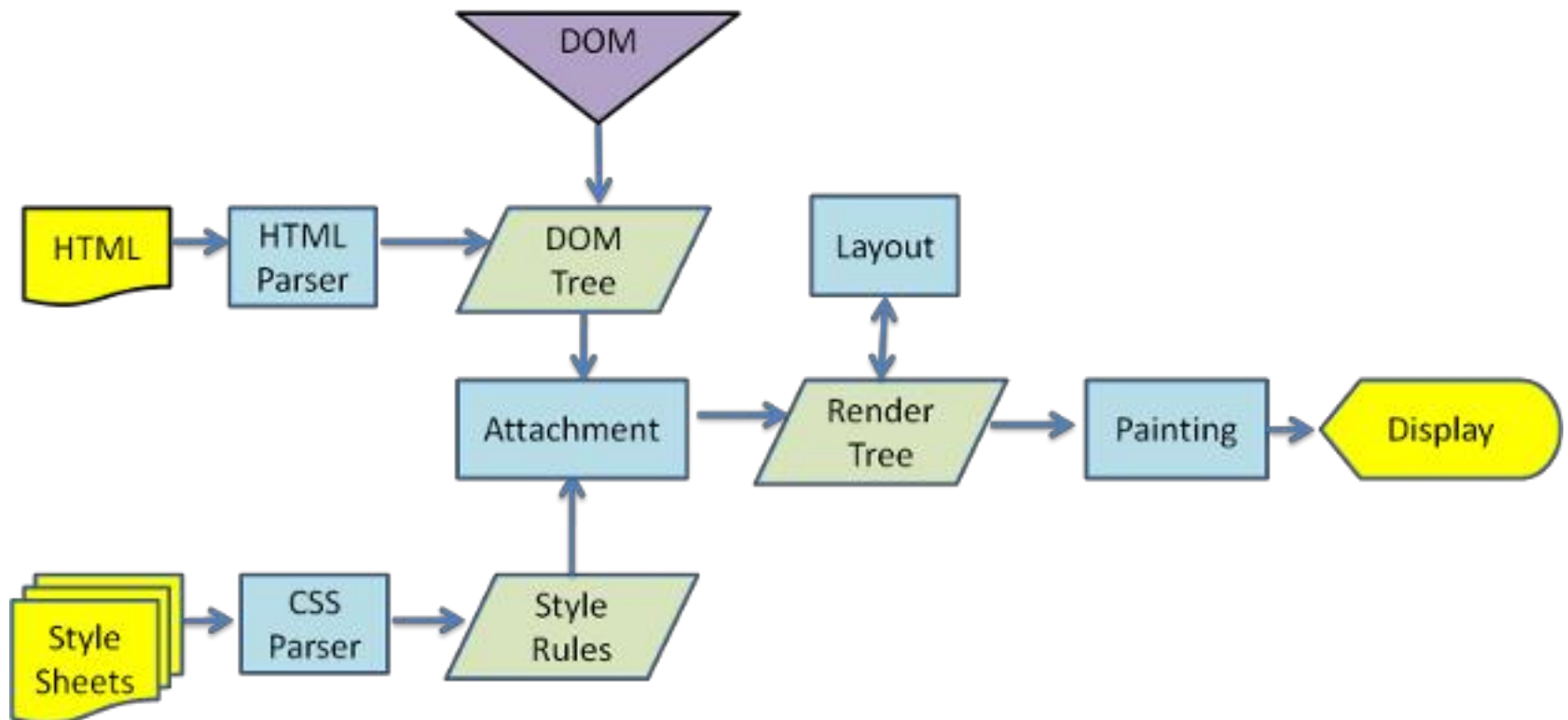
body {
  width: 50%;
}

div {
  height: 200px;
}

p {
  width: 50%;
}

span {
  position: absolute;
  right: 25px;
}
```

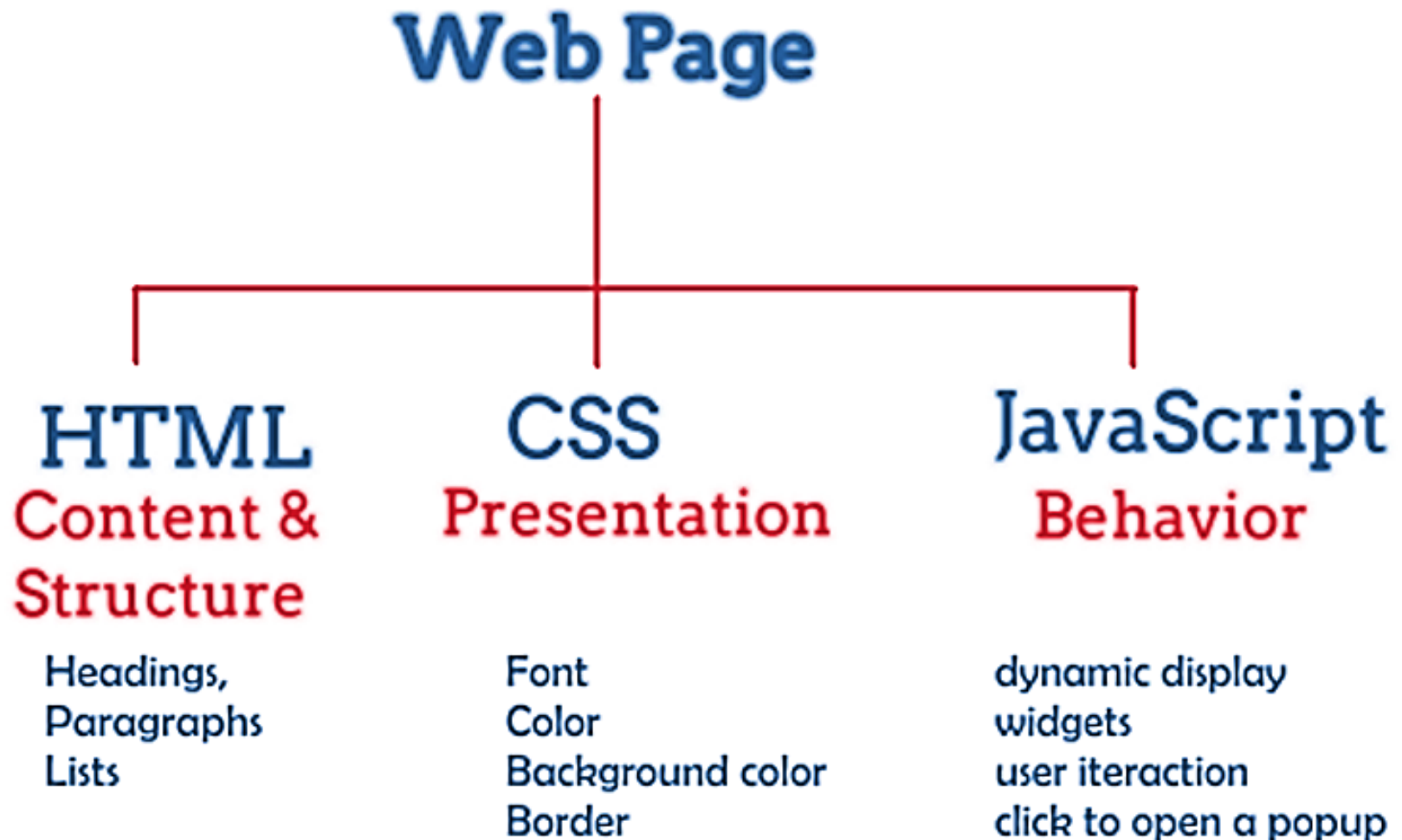
7. การทำงานของ Web Browser



8. JavaScript



Html, CSS, JavaScript



Html, CSS, JavaScript



HTML



HTML + CSS



HTML + CSS
+ JAVASCRIPT



house folder



index.html



styles.css



scripts.js

8. JavaScript

- เดิมทีภาษา **JavaScript** ทำงานที่ **Web browser** เป็นหลัก โดยทำหน้าที่เปลี่ยนแปลงโครงสร้างของ **HTML** เป็นหลัก



- JavaScript Can Change HTML Content
- JavaScript Can Change HTML Attribute Values
- JavaScript Can Change HTML Styles (CSS)
- JavaScript Can Hide/Show HTML Elements

8. JavaScript

- **JavaScript Can Change HTML Content**

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>What Can JavaScript Do?</h2>
```

```
<p id="demo">JavaScript can change HTML content.</p>
```

```
<button type="button" onclick="document.getElementById('demo').innerHTML  
= 'Hello JavaScript!'">Click Me!</button>
```

```
</body>
```

```
</html>
```

What Can JavaScript Do?

JavaScript can change HTML content.

Click Me!

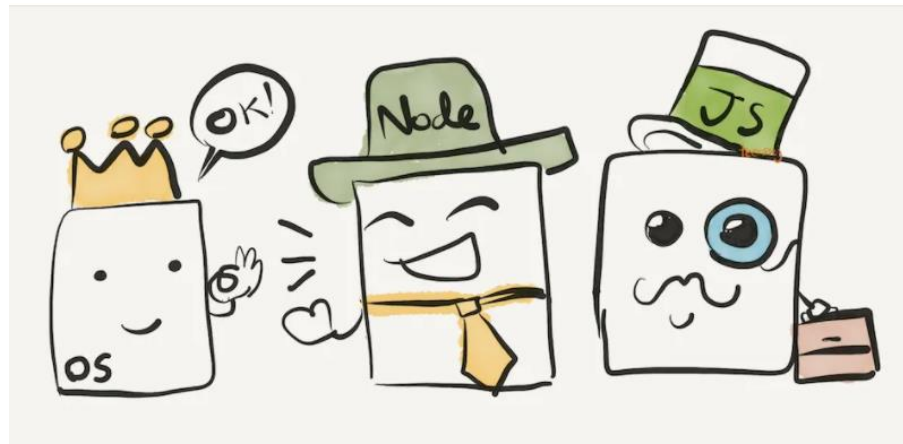
What Can JavaScript Do?

Hello JavaScript!

Click Me!

8. JavaScript

- ต่อมา ผู้คนอยากให้ **JavaScript** ทำงานนอก **Web Browser** ได้ เพื่อเพิ่มความสามารถของมัน จึงมีคนคิดค้นสร้างสิ่งที่เรียกว่า **Node**
- **Node** หรือ **Node.js** คือ **JavaScript Runtime** ที่ถูกสร้างด้วย **Chrome V8 JavaScript Engine**
- โดย **JavaScript Runtime** ทำหน้าที่เพื่อ อ่าน **code** ภาษา **javascript** และทำงานตามคำสั่ง เพื่อให้ **OS** ต่างๆ ทำงานได้



Full Stack JavaScript Tools and Technologies

Front End



Or



React

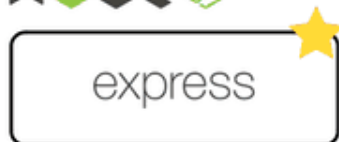


Knockout.



BACKBONE.JS

node.js +



Or

METEOR

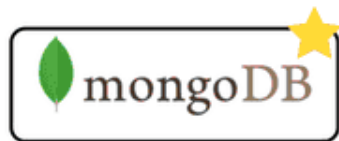
sails

restify



KeystoneJS

Database



Or

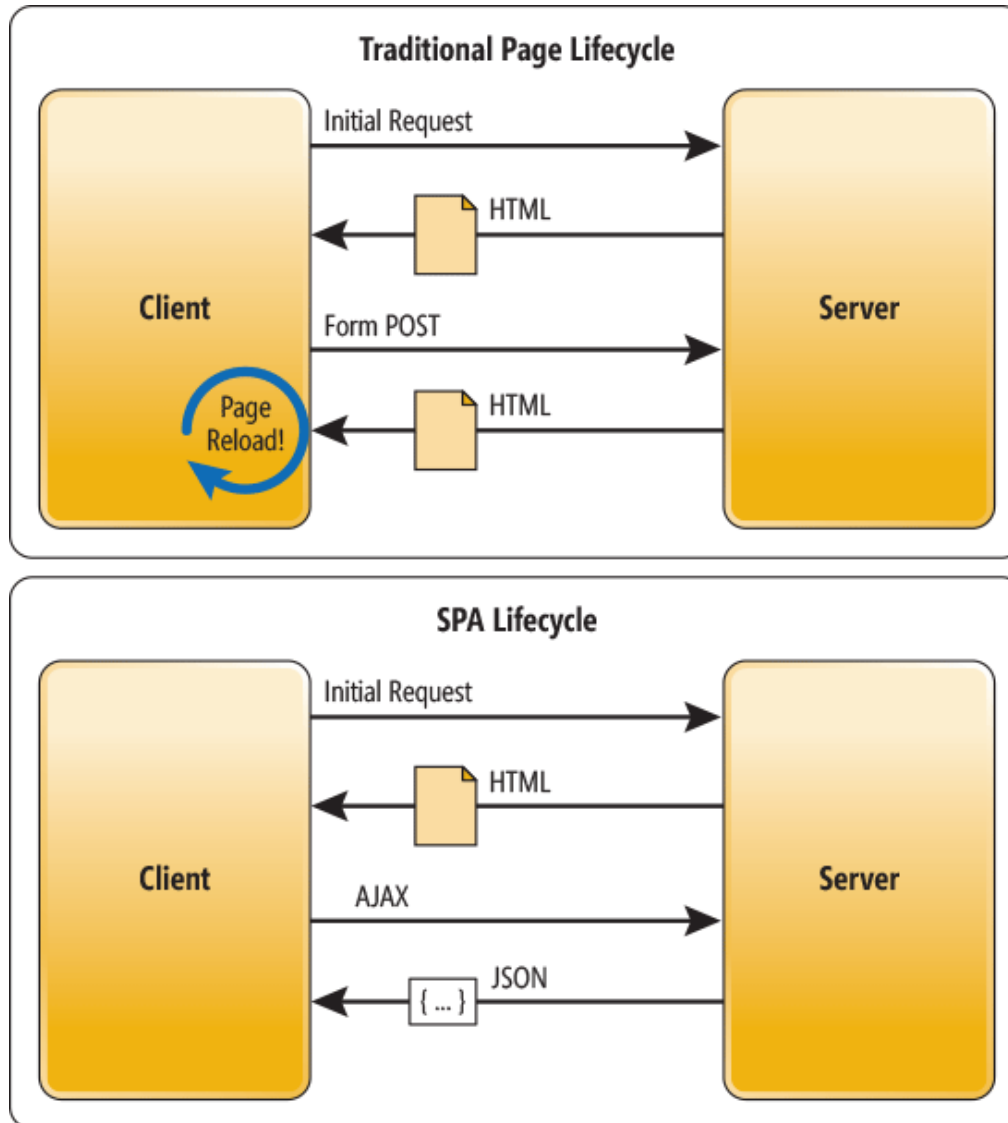
MySQL

PostgreSQL

CouchDB

cassandra

9. Single Page Application

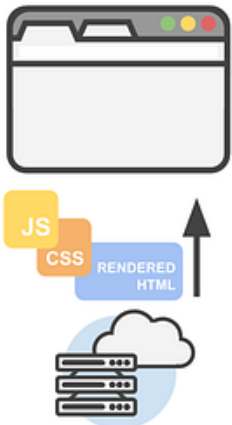


Single Page Application

Server-Side Rendering (SSR)

Step 1

The page loads and requests static assets from the web server. The data is included in the HTML.



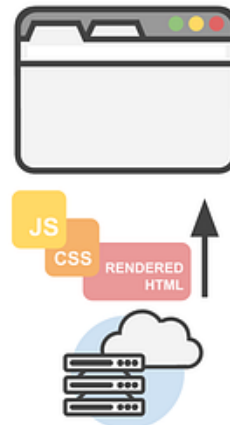
Step 2

The browser has all the information it needs, the page is rendered immediately.



Step 3

The user interacts with the page. This triggers a new request to the server, the page goes blank until it reloads.



Step 4

The new page loads, showing new content to the user.

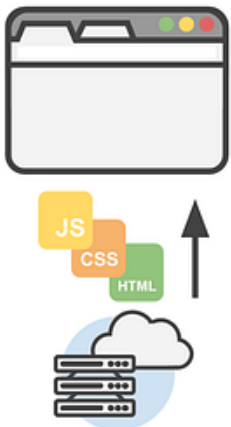


Single Page Application

Single-Page Application (SPA)

Step 1

The page loads and requests static assets from the web server.



Step 2

The frontend framework renders the layout using the HTML and CSS.
The JavaScript is executed and requests data.



Step 3

The JavaScript loads the data in the page.



Step 4

The user interacts with the page, more data is requested and loaded.
The page is updated without having to reload.



Single Page Application

Companies Using Single-page Applications



Twitter



Instagram



Airbnb



Facebook



Netflix



Gmail



Uber



10. Introduction to API

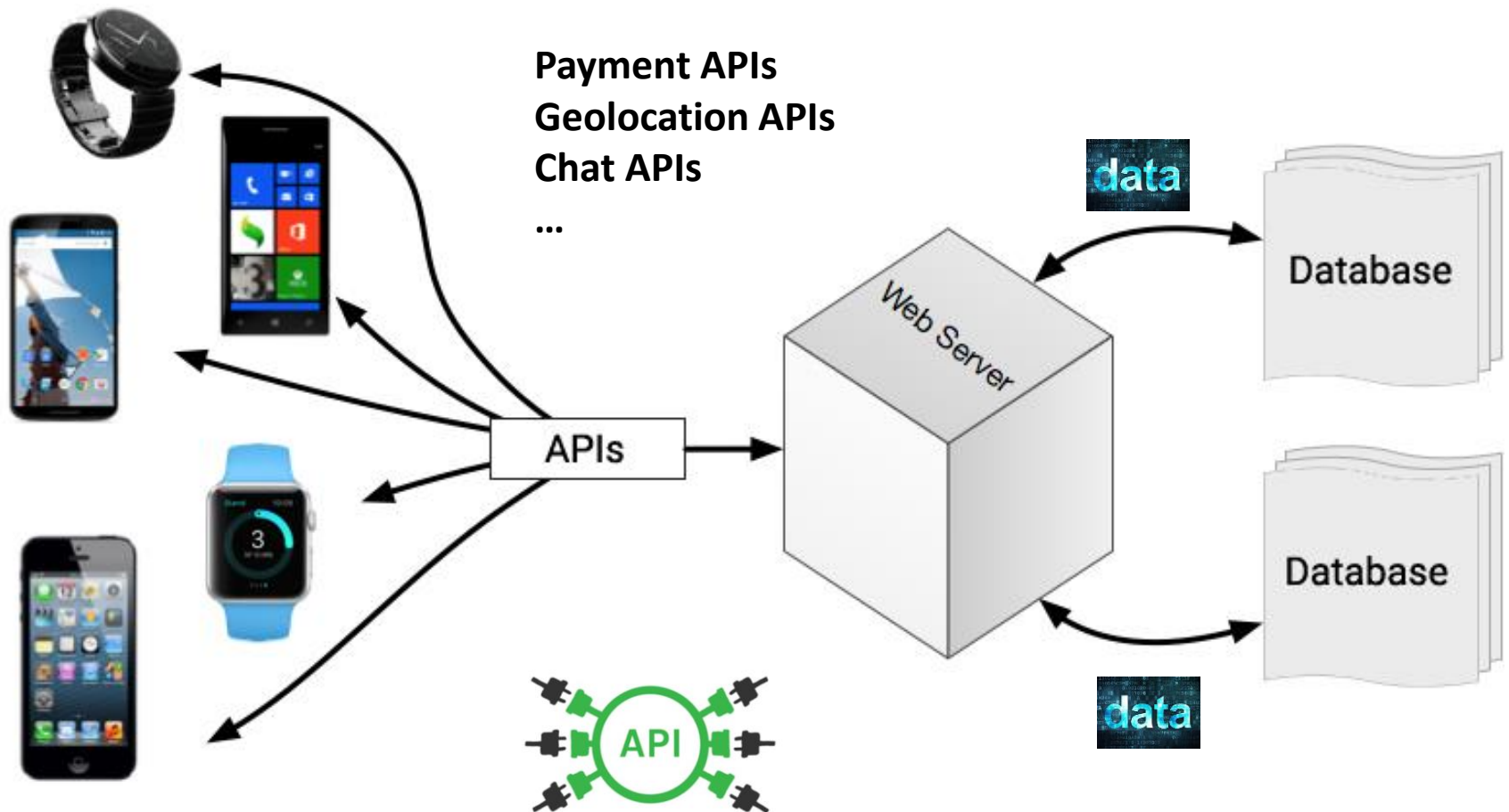


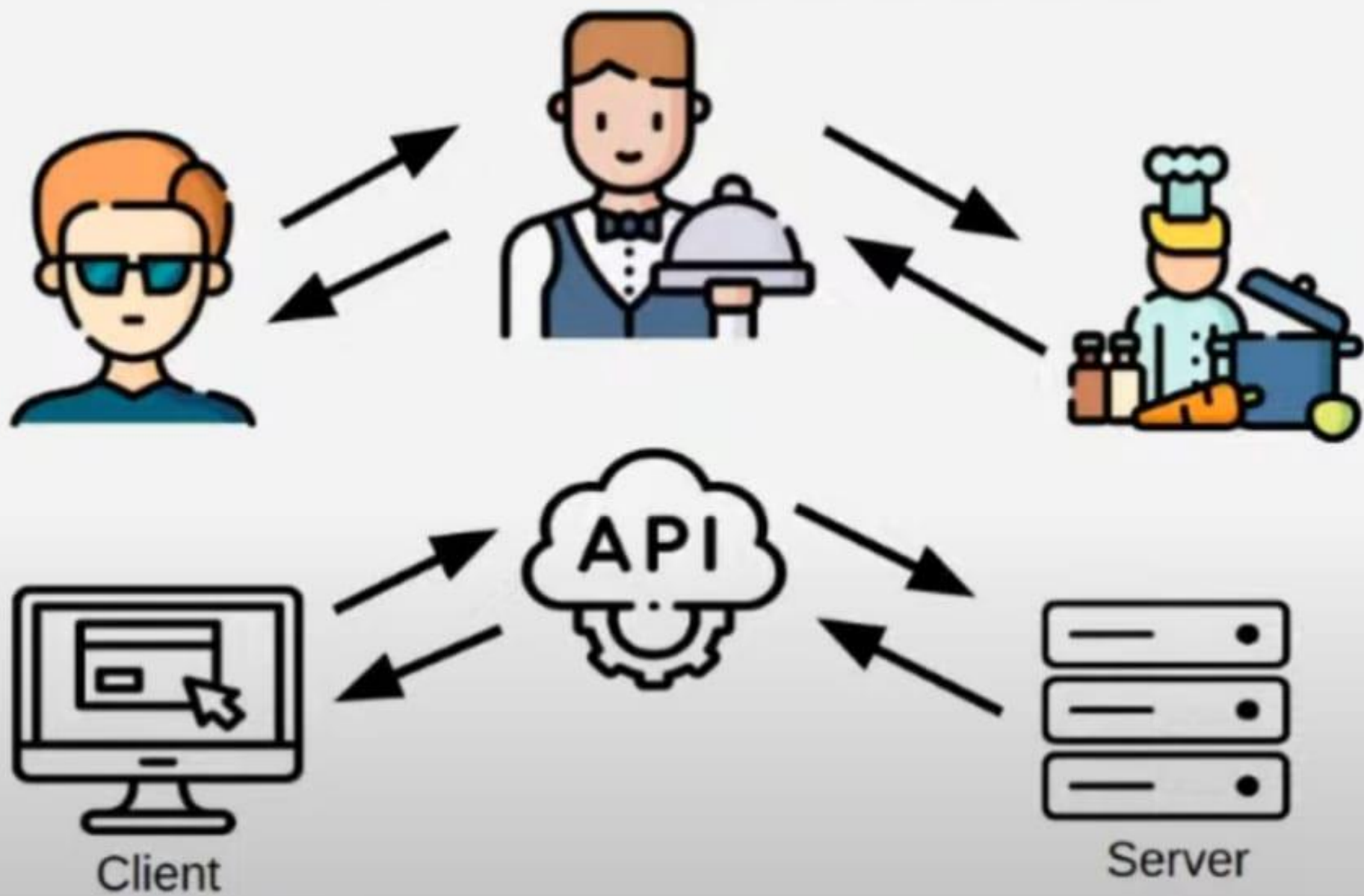
โรงไฟฟ้า ผลิตไฟฟ้า เพื่อให้บริการไฟฟ้า
คำถาม ถ้าไม่มีปลั๊กตัวเมียมาให้ เราจะใช้ไฟฟ้า อย่างไร ?



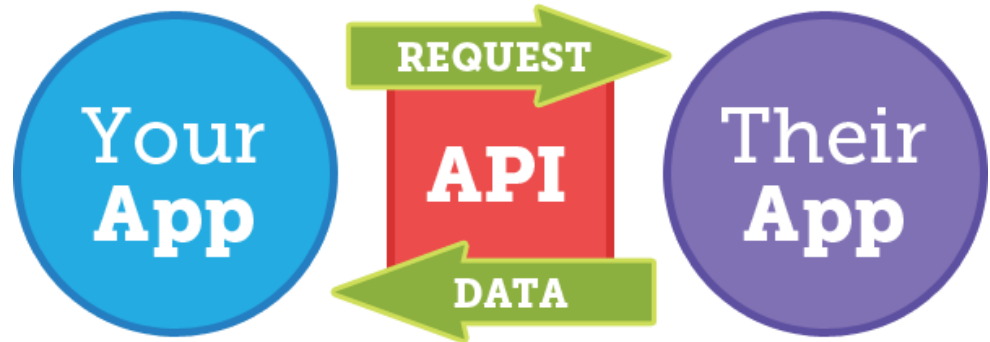
10. Introduction to API

API ย่อมาจาก Application Programming Interface





An API allows one piece of software talk to another.



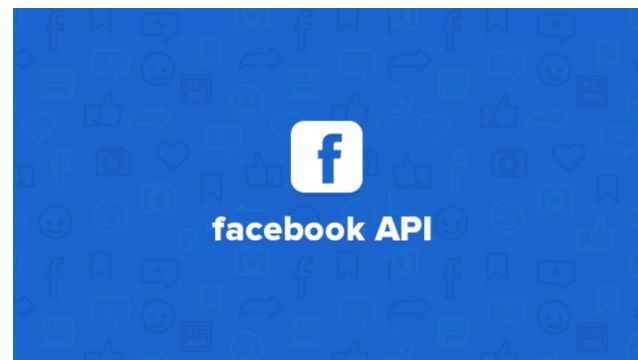
ตัวอย่าง API

Google Nearby API



Google Maps APIs

YouTube
Data API v3



ตัวอย่าง API ให้ข้อมูล อัตราแลกเปลี่ยน

<https://api.exchangeratesapi.io/latest>

← → ↻ 🔒 <https://api.exchangeratesapi.io/latest>

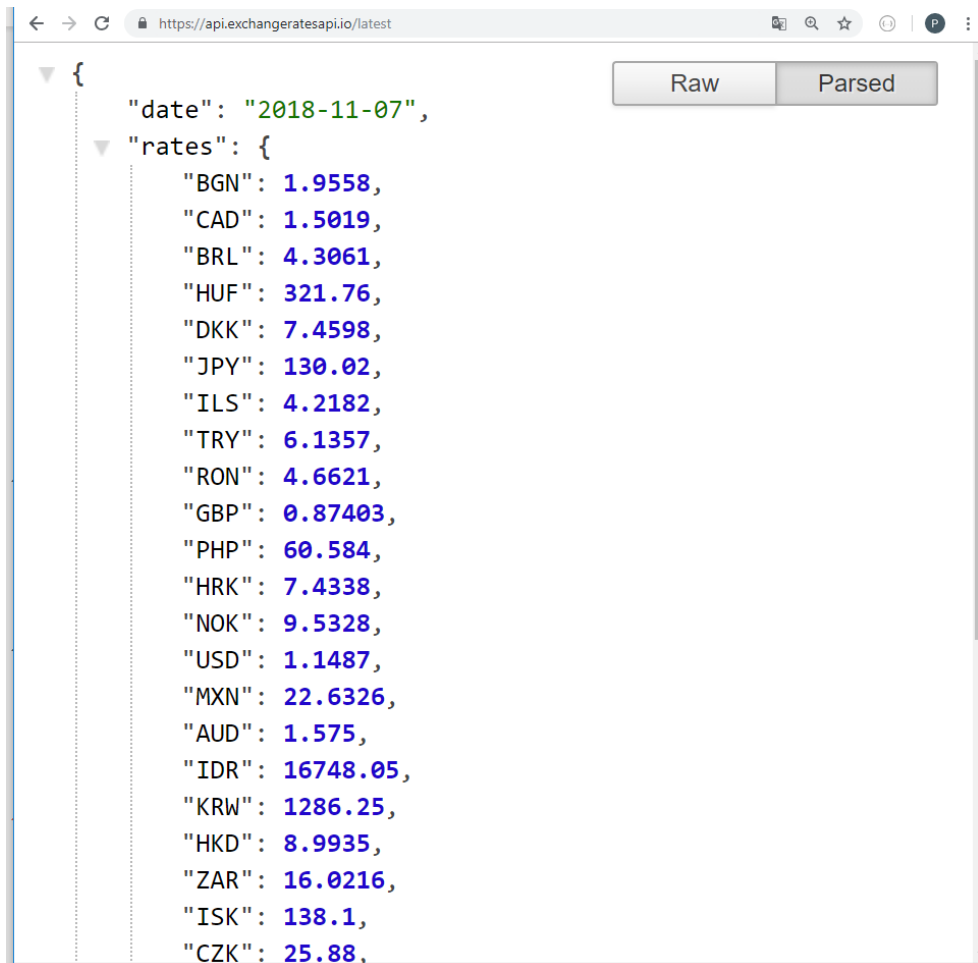
🔍 ☆ P ⋮

```
{"date": "2018-11-07", "rates":  
{ "BGN": 1.9558, "CAD": 1.5019, "BRL": 4.3061, "HUF": 321.76, "DKK": 7.4598,  
  "JPY": 130.02, "ILS": 4.2182, "TRY": 6.1357, "RON": 4.6621, "GBP": 0.87403,  
  "PHP": 60.584, "HRK": 7.4338, "NOK": 9.5328, "USD": 1.1487, "MXN": 22.6326,  
  "AUD": 1.575, "IDR": 16748.05, "KRW": 1286.25, "HKD": 8.9935, "ZAR": 16.0216,  
  "ISK": 138.1, "CZK": 25.88, "THB": 37.66, "MYR": 4.7734, "NZD": 1.6954,  
  "PLN": 4.294, "SEK": 10.3313, "RUB": 75.7709, "CNY": 7.9426, "SGD": 1.5729,  
  "CHF": 1.1444, "INR": 83.28}, "base": "EUR" }
```


ตัวอย่าง API ให้ข้อมูล อัตราแลกเปลี่ยน

<https://api.exchangeratesapi.io/latest>

ติดตั้ง plugin **json formatter**



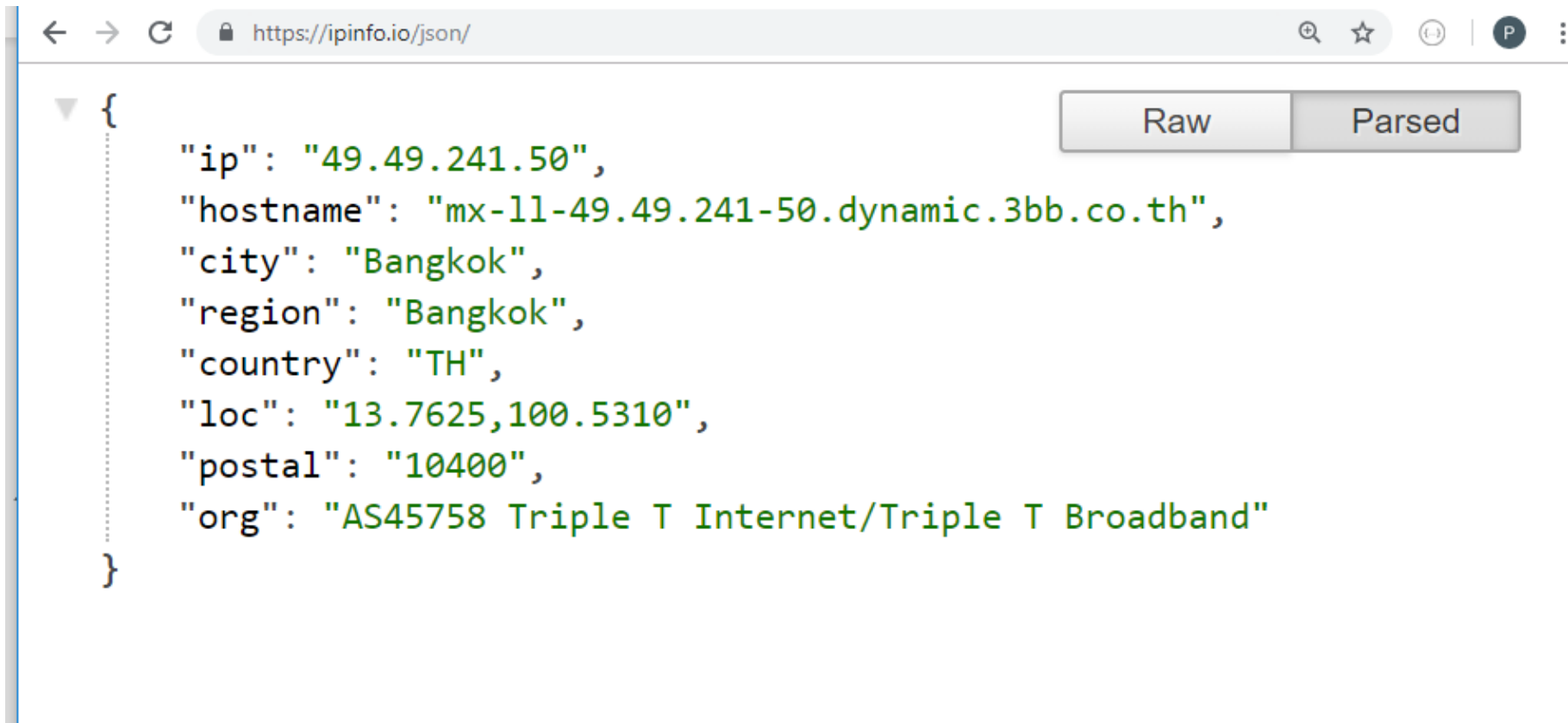
The screenshot shows a web browser window with the URL <https://api.exchangeratesapi.io/latest>. The response is displayed as a JSON object, which has been formatted by a plugin. The JSON structure is as follows:

```
{
  "date": "2018-11-07",
  "rates": {
    "BGN": 1.9558,
    "CAD": 1.5019,
    "BRL": 4.3061,
    "HUF": 321.76,
    "DKK": 7.4598,
    "JPY": 130.02,
    "ILS": 4.2182,
    "TRY": 6.1357,
    "RON": 4.6621,
    "GBP": 0.87403,
    "PHP": 60.584,
    "HRK": 7.4338,
    "NOK": 9.5328,
    "USD": 1.1487,
    "MXN": 22.6326,
    "AUD": 1.575,
    "IDR": 16748.05,
    "KRW": 1286.25,
    "HKD": 8.9935,
    "ZAR": 16.0216,
    "ISK": 138.1,
    "CZK": 25.88,
  }
}
```

The browser window also shows a 'Raw' button and a 'Parsed' button, indicating that the JSON has been successfully parsed.

ตัวอย่าง API ให้ข้อมูล IP Address

<https://ipinfo.io/json/>

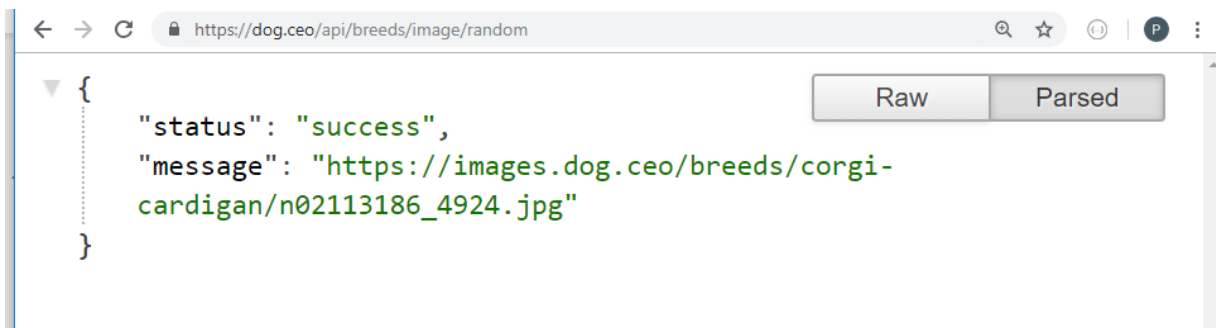
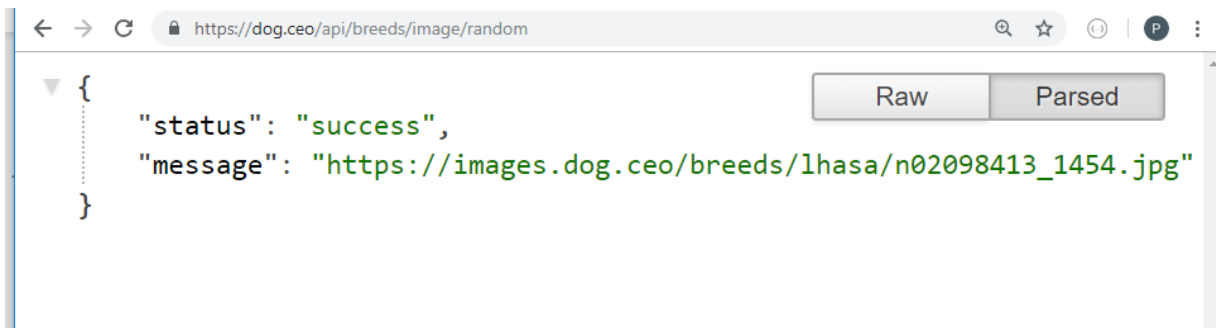


The screenshot shows a web browser window with the address bar displaying <https://ipinfo.io/json/>. The page content displays a JSON object representing IP information. To the right of the JSON object are two buttons: "Raw" and "Parsed".

```
{  
  "ip": "49.49.241.50",  
  "hostname": "mx-11-49.49.241-50.dynamic.3bb.co.th",  
  "city": "Bangkok",  
  "region": "Bangkok",  
  "country": "TH",  
  "loc": "13.7625,100.5310",  
  "postal": "10400",  
  "org": "AS45758 Triple T Internet/Triple T Broadband"  
}
```

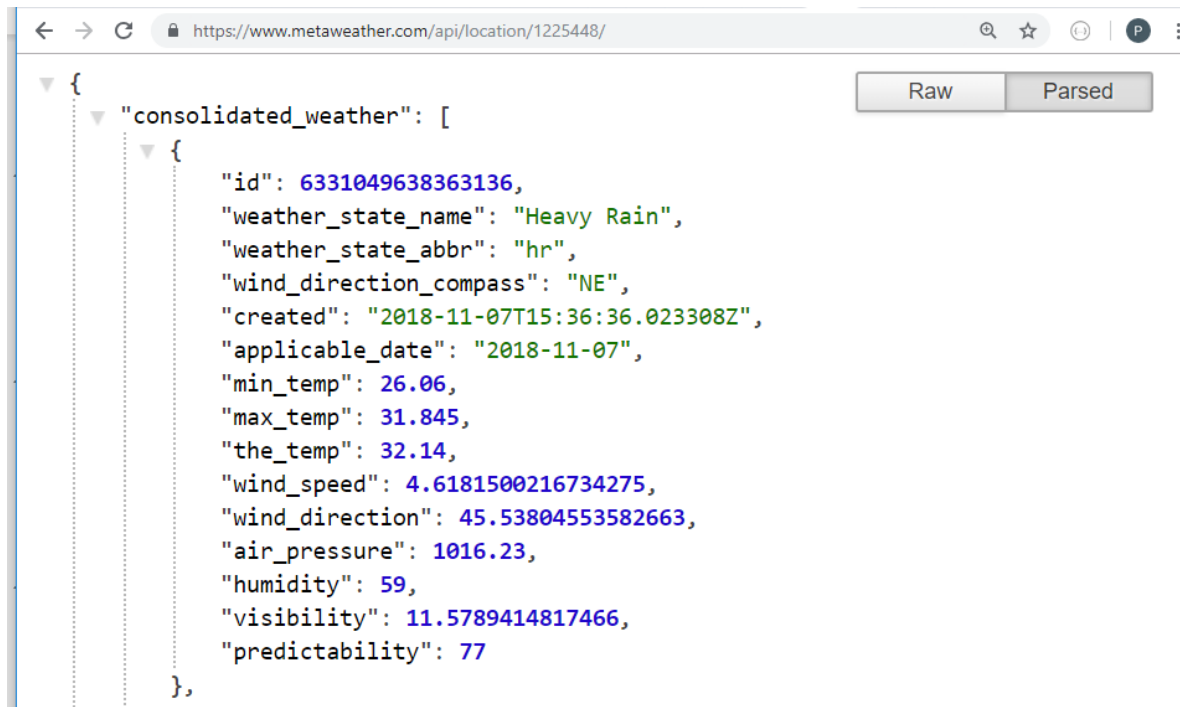
ตัวอย่าง API ให้ข้อมูล Random ภาพสุนัข

<https://dog.ceo/api/breeds/image/random>



ตัวอย่าง API ให้ข้อมูล สภาพอากาศ

<https://www.metaweather.com/api/location/1225448/>



```
{
  "consolidated_weather": [
    {
      "id": 6331049638363136,
      "weather_state_name": "Heavy Rain",
      "weather_state_abbr": "hr",
      "wind_direction_compass": "NE",
      "created": "2018-11-07T15:36:36.023308Z",
      "applicable_date": "2018-11-07",
      "min_temp": 26.06,
      "max_temp": 31.845,
      "the_temp": 32.14,
      "wind_speed": 4.6181500216734275,
      "wind_direction": 45.53804553582663,
      "air_pressure": 1016.23,
      "humidity": 59,
      "visibility": 11.5789414817466,
      "predictability": 77
    }
  ]
}
```

ตัวอย่าง API อื่นๆ สามารถดูได้จาก

<https://github.com/toddmotto/public-apis> หรือค้นจาก google

การเขียน html ดึงค่าข้อมูล ผ่าน API

<https://api.exchangeratesapi.io/latest>

ไฟล์ index.html

```
File Edit Format View Help
<html>
<head>
  <title>Welcome to JSON</title>
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.3.1/jquery.min.js"></script>
</head>
<body>
  <script type="text/javascript">
    $.getJSON('https://api.exchangeratesapi.io/latest', function(response) {
      document.getElementById("data").innerHTML = response.rates.THB
      document.getElementById("data1").innerHTML = response.date
    });
  </script>
  <div style = "color:red" id="data" > </div>
  <div style = "color:blue" id="data1"> </div>
</body>
</html>
```

Request

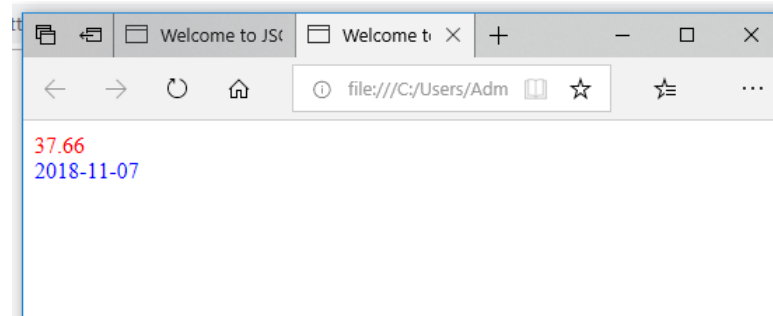
Data

```
https://api.exchangeratesapi.io/latest
{
  "PHP": 60.584,
  "HRK": 7.4338,
  "NOK": 9.5328,
  "USD": 1.1487,
  "MXN": 22.6326,
  "AUD": 1.575,
  "IDR": 16748.05,
  "KRW": 1286.25,
  "HKD": 8.9935,
  "ZAR": 16.0216,
  "ISK": 138.1,
  "CZK": 25.88,
  "THB": 37.66,
  "MYR": 4.7734,
  "NZD": 1.6954,
  "PLN": 4.294,
  "SEK": 10.3313,
  "RUB": 75.7709,
  "CNY": 7.9426,
  "SGD": 1.5729,
  "CHF": 1.1444,
  "INR": 83.28
},
"base": "EUR"
```

เปิดไฟล์ผ่าน Web browser



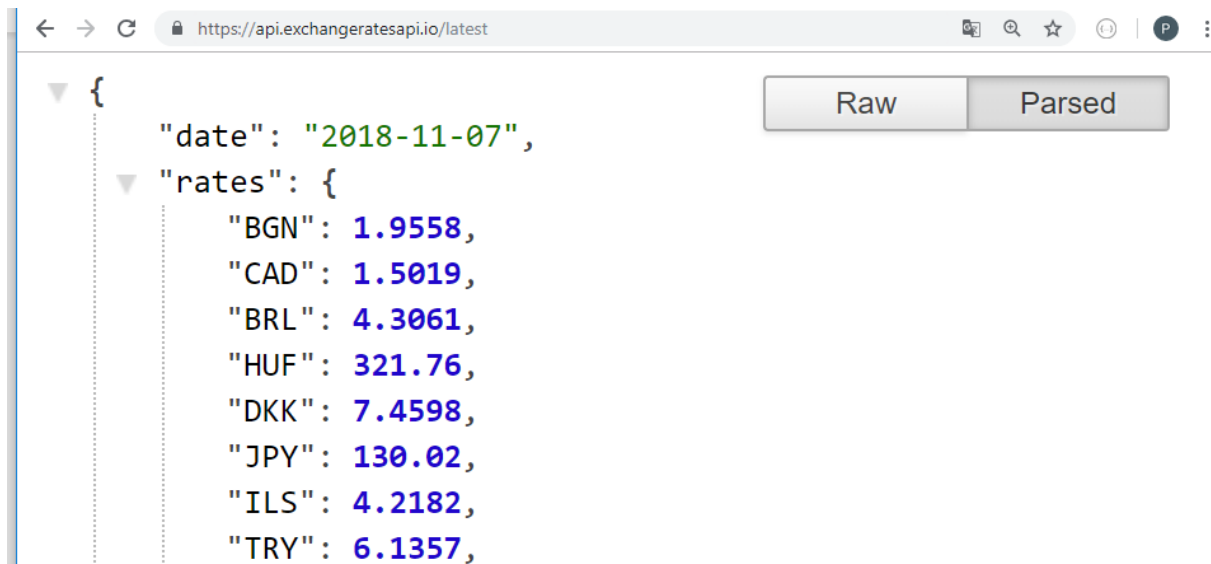
แสดงผล



9.1 REST API

REST หรือ Representational State Transfer เป็นวิธีการสร้าง Web Service รูปแบบหนึ่งที่อาศัย HTTP Method (GET, POST, PUT, DELETE) ในการทำงาน และส่งผลกลับมาในรูปแบบของ JSON หรือ XML

เช่น <https://api.exchangeratesapi.io/latest>



REST API

ตัวอย่าง API Document

workshop\$Order Entity CRUD operations



GET **/entities/workshop\$Order** Gets a list of entities: workshop\$Order

POST **/entities/workshop\$Order** Creates new entity: workshop\$Order

GET **/entities/workshop\$Order/{entityId}** Gets a single entity by identifier: workshop\$Order

PUT **/entities/workshop\$Order/{entityId}** Updates the entity: workshop\$Order

DELETE **/entities/workshop\$Order/{entityId}** Deletes the entity: workshop\$Order

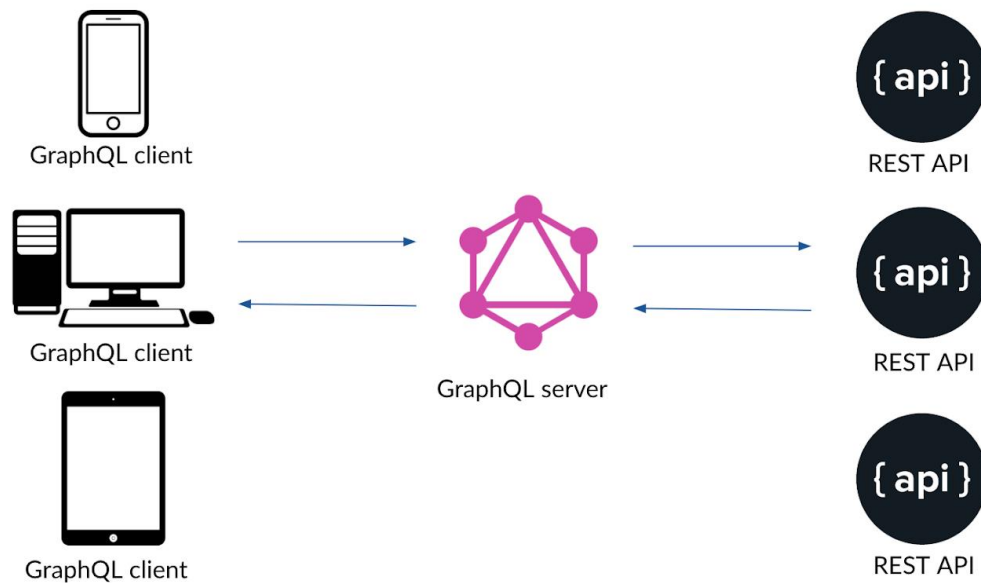
GET **/entities/workshop\$Order/search** Find entities by filter conditions: workshop\$Order

POST **/entities/workshop\$Order/search** Find entities by filter conditions: workshop\$Order

11. GraphQL

- เนื่องจาก การดึงค่าแบบ **Rest API** บางครั้งข้อมูลมากเกินไปจนจำเป็น บางครั้งต้องดึงหลายครั้ง เพื่อนำข้อมูลมาประมวลผล หรือเอามาประกอบกัน จึงทำให้เกิดแนวคิด ของ **GraphQL**

“ GraphQL is a query language for APIs and a runtime for fulfilling those queries with your existing data. ”



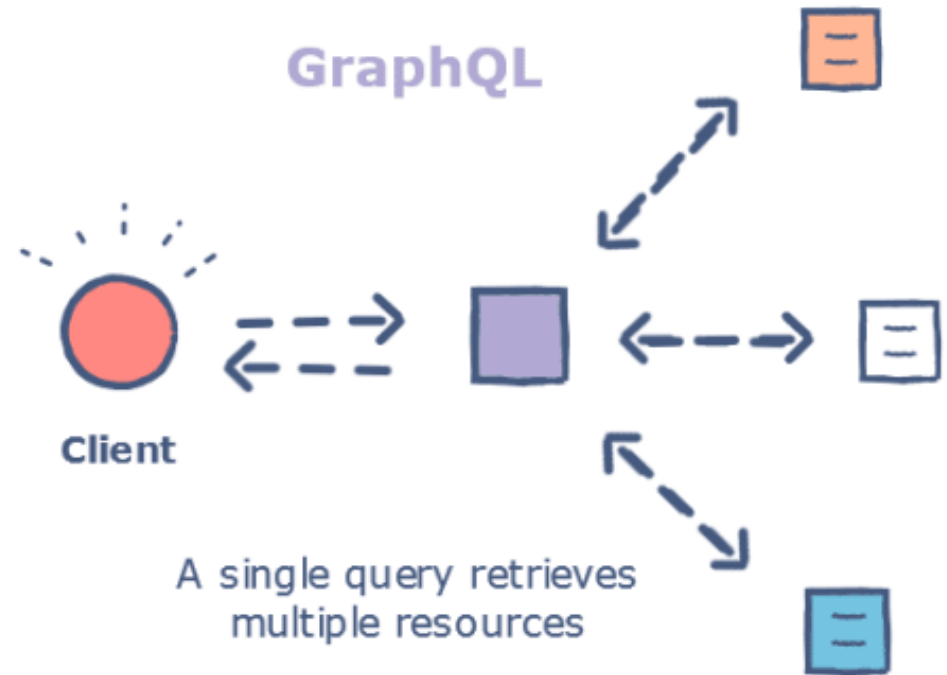
GraphQL

REST



Retrieving multiple

GraphQL



A single query retrieves multiple resources

ขอบคุณครับ

ขอให้โชคดีในการสอบครับ เป็นกำลังใจให้ทุก ๆ คนครับ
สิ่งใดพลาดไป ขออภัย และขอโทษมา ณ ที่นี้ด้วยครับ